



Faculty of Technology and Society
Department of Computer Science and Media Technology

Master Thesis Project, Spring 2018

**A performance measurement of a Speaker
Verification system based on a variance in data
collection for Gaussian Mixture Model and
Universal Background Model**

By

B.S.E. in Computer Engineering Zeid Bekli
B.S.E.in Computer Engineering William Ouda

Supervisors:

Arezoo Sarkheyli Haegele

Examiner:

Johan Holmgren

Contact information

Authors:

Zeid Bekli

E-mail: zeidbekli48@gmail.com

William Ouda

E-mail: tative21@gmail.com

Supervisor:

Arezoo Sarkheyli Haegele

E-mail: arezoo.sarkheyli-haegele@mah.se

Malmö University, Department of Computer Science and Media Technology.

Examiner:

Johan Holmgren

E-mail: johan.holmgren@mau.se

Malmö University, Department of Computer Science and Media Technology.

Abstract

Voice recognition has become a more focused and researched field in the last century, and new techniques to identify speech has been introduced. A part of voice recognition is speaker verification which is divided into Front-end and Back-end. The first component is the front-end or feature extraction where techniques such as Mel-Frequency Cepstrum Coefficients (MFCC) is used to extract the speaker specific features of a speech signal, MFCC is mostly used because it is based on the known variations of the humans ear's critical frequency bandwidth. The second component is the back-end and handles the speaker modeling. The back-end is based on the Gaussian Mixture Model (GMM) and Gaussian Mixture Model-Universal Background Model (GMM-UBM) methods for enrollment and verification of the specific speaker. In addition, normalization techniques such as Cepstral Means Subtraction (CMS) and feature warping is also used for robustness against noise and distortion. In this paper, we are going to build a speaker verification system and experiment with a variance in the amount of training data for the true speaker model, and to evaluate the system performance. And further investigate the area of security in a speaker verification system then two methods are compared (GMM and GMM-UBM) to experiment on which is more secure depending on the amount of training data available.

This research will therefore give a contribution to how much data is really necessary for a secure system where the False Positive is as close to zero as possible, how will the amount of training data affect the False Negative (FN), and how does this differ between GMM and GMM-UBM.

The result shows that an increase in speaker specific training data will increase the performance of the system. However, too much training data has been proven to be unnecessary because the performance of the system will eventually reach its highest point and in this case it was around 48 min of data, and the results also show that the GMM-UBM model containing 48- to 60 minutes outperformed the GMM models.

Keywords: *Speaker recognition; Speaker Verification; Speaker authentication; Speaker classification Text-dependent; Text-independent; Gaussian mixture model; Gaussian Mixture Model-Universal Background Model; Biometric System; Equal Error Rate; False Negative Rate; False Positive Rate; Mel Frequency Cepstrum Coefficients;*

Acknowledgement

We would like to show our appreciation to our supervisors Christoffer Cronström, Niklas Hjern and Arezoo Sarkheyli Haegele for taking the time to guide and support us during this thesis.

Table of contents

1. Introduction	1
1.1 Research Motivation	3
1.2 Research Question	4
1.3 Limitation	4
2. Theoretical Background	6
2.1 Biometrics System	6
2.1.1 Verification System	7
2.1.2 Identification System	7
2.1.3 Speaker Verification System	8
2.1.4 Performance Metrics for Biometric Systems	8
2.2 Digital Signal Processing	9
2.2.1 Frame-Blocking and Frame-Overlapping	9
2.2.2 Hamming Window	9
2.2.3 Discrete Fourier Transform	10
2.2.4 Spectrum	10
2.2.5 Cepstrum	10
2.2.6 Mel-Frequency Cepstrum Coefficients	10
2.3 Speech Signals	11
2.4 Voice Modeling	11
2.4.1 Gaussian Mixture Model	12
2.4.2 Universal Background Model	13
3. Related Work	15
3.1 Hossan. M.A., et al. - A Novel Approach for MFCC Feature Extraction	15
3.2 Fang, Z., et al. - Comparison of Different Implementations of MFCC	16
3.3 Chao-Yu, L., et al. - User Identification Design by Fusion of Face Recognition and Speaker Recognition	16
3.4 Mohamed, K.O., et al. - Training Universal Background Models for Speaker Recognition	17
3.5 Taufiq, H., et al. - A Study on Universal Background Model Training in Speaker Verification	18
3.6 Pelecanos, J., et al. - Feature Warping for Robust Speaker Verification	19
3.7 Kumari, T. J., et al. - Comparison of LPCC and MFCC features and GMM and GMM-UBM modeling for limited data speaker verification	20
4. Method	21
4.1 Construct a Conceptual Framework	22
4.2 Develop a System Architecture	22

4.3 Analyze and Design the System.....	23
4.4 Build the (prototype) System.....	24
4.5 Observe and Evaluate the System.....	24
5. Results.....	25
5.1 The System.....	25
5.2 The Feature Extraction Phase	26
5.2.1 Mel-Frequency Cepstrum Coefficients.....	26
5.2.1.1 Sampling.....	26
5.2.1.2 Pre-Emphasis	26
5.2.1.3 Frame Blocking.....	29
5.2.1.4 Hamming Window	29
5.2.1.5 Discrete Fourier Transform.....	30
5.2.1.6 Mel-Filterbank	31
5.2.1.7 Discrete Cosine Transform	32
5.2.2 Normalization and Derivatives.....	33
5.3 The Enrollment Phase.....	35
5.4 The Verification Phase	35
5.5 Results.....	36
5.5.1 The GMM-UBM Method	37
5.5.2 The GMM Method.....	43
5.6 Analysis	49
6 Discussion	52
6.1 Method Discussion	52
6.2 The Speaker Verification System Components	52
6.3 Speaker Verification System Performance	53
7 Conclusion	55
7.1 Answering the Research Question.....	55
7.1.1 How does the GMM-UBM speaker verification system perform based on different amount of data collection for the true speaker model?	55
7.1.2 How does the GMM speaker verification system perform compared to the GMM-UBM, based on different amount of data collection for the true speaker model?.....	56
7.2 Further Work.....	56
7.3 Contribution of this Thesis	56
References	57

LIST OF FIGURES

FIGURE 1: THE OPERATIONS OF A TYPICAL BIOMETRIC SYSTEM [1]	6
FIGURE 2: THE VERIFICATION PROCESS	7
FIGURE 3: IDENTIFICATION PHASE WITH A SET OF SPEAKERS.....	7
FIGURE 4: EER METRIC STRUCTURE [40]	8
FIGURE 5: BLOCK DIAGRAM OF MFCC WITH DELTA AND DELTA-DELTA	11
FIGURE 6: THE PROCESS OF THE GMM-UBM	14
FIGURE 7: SYSTEMS DEVELOPMENT RESEARCH METHODOLOGY[33].....	21
FIGURE 8: A COMMON SYSTEM ARCHITECTURE FOR SPEAKER VERIFICATION.....	22
FIGURE 9: SYSTEM ARCHITECTURE AND COMPONENTS FOR A SPEAKER VERIFICATION SYSTEM[34]	23
FIGURE 10: THE THREE MAIN COMPONENTS FOR A SPEAKER VERIFICATION SYSTEM.....	25
FIGURE 11A: ORIGINAL SIGNAL.....	27
FIGURE 11B: PRE-EMPHASIS SIGNAL.....	27
FIGURE 11C: ORIGINAL SIGNAL (PS).....	28
FIGURE 11D: PRE-EMPHASIS SIGNAL (PS)	28
FIGURE 12A: A FRAME FROM THE SPEECH SIGNAL.....	29
FIGURE 12B: HAMMING WINDOW ON THE FRAME.....	30
FIGURE 13A: THE FILTER BANK WITH 32 FILTERS.....	31
FIGURE 13B: THE ENERGIES EXTRACTED FROM A SINGLE FRAME	32
FIGURE 14A: ORIGINAL DISTRIBUTION OF COEFFICIENT	33
FIGURE 14B: NORMALIZED DISTRIBUTION OF COEFFICIENT.....	34
FIGURE 15A: SHOWS THE SCORE DISTRIBUTION FOR MODEL ONE BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER ONE).....	39
FIGURE 15B: SHOWS THE SCORE DISTRIBUTION FOR MODEL EIGHT BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER TWO).....	39
FIGURE 15C: SHOWS THE SCORE DISTRIBUTION FOR MODEL ONE BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER TWO).....	40
FIGURE 15D: SHOWS THE SCORE DISTRIBUTION FOR MODEL EIGHT BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER TWO).....	40
FIGURE 16A: SHOWS THE DET CURVE FOR MODEL ONE (SPEAKER ONE).	41
FIGURE 16B: SHOWS THE DET CURVE FOR MODEL EIGHT (SPEAKER ONE).	41
FIGURE 16C: SHOWS THE DET CURVE FOR MODEL ONE (SPEAKER TWO).....	42
FIGURE 16D: SHOWS THE DET CURVE FOR MODEL EIGHT (SPEAKER TWO).	42
FIGURE 17A: SHOWS THE SCORE DISTRIBUTION FOR MODEL ONE BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER ONE).....	45
FIGURE 17B: SHOWS THE SCORE DISTRIBUTION FOR MODEL EIGHT BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER ONE).....	45

FIGURE 17C: SHOWS THE SCORE DISTRIBUTION FOR MODEL ONE BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER(SPEAKER TWO).	46
FIGURE 17D: SHOWS THE SCORE DISTRIBUTION FOR MODEL EIGHT BETWEEN THE TRUE SPEAKER AND IMPOSTER SPEAKER (SPEAKER TWO).	46
FIGURE 18A: SHOWS THE DET CURVE FOR MODEL ONE (SPEAKER ONE).	47
FIGURE 18B: SHOWS THE DET CURVE FOR MODEL EIGHT (SPEAKER ONE).	47
FIGURE 18C: SHOWS THE DET CURVE FOR MODEL ONE (SPEAKER TWO).	48
FIGURE 18D: SHOWS THE DET CURVE FOR MODEL EIGHT(SPEAKER TWO).	48
FIGURE 19A: THE EER FOR THE EIGHT MODELS FOR GMM AND GMM-UBM (SPEAKER ONE).	50
FIGURE 19A: THE EER FOR THE EIGHT MODELS FOR GMM AND GMM-UBM (SPEAKER TWO).	50
FIGURE 20A: THE FN FOR THE EIGHT MODELS FOR GMM AND GMM-UBM (SPEAKER ONE).	50
FIGURE 20B: THE FN FOR THE EIGHT MODELS FOR GMM AND GMM-UBM (SPEAKER TWO).	51

LIST OF TABLES

TABLE 1: THE MAIN CONTRIBUTION OF THIS RESEARCH COMPARED TO RELATED WORK 4

TABLE 2: RESULTS FOR SPEAKER ONE BASED ON EIGHT MODELS WITH DIFFERENT AMOUNT TRAINING DATA
WITH GMM-UMB MODELING. 37

TABLE 3: RESULTS FOR SPEAKER TWO BASED ON EIGHT MODELS WITH DIFFERENT AMOUNT TRAINING DATA
WITH GMM-UMB MODELLING. 38

TABLE 4: RESULTS FOR SPEAKER ONE BASED ON EIGHT MODELS WITH DIFFERENT AMOUNT TRAINING DATA
WITH GMM MODELING. 43

TABLE 5: RESULTS FOR SPEAKER ONE BASED ON EIGHT MODELS WITH DIFFERENT AMOUNT TRAINING DATA
WITH GMM MODELING. 44

List of acronyms

CMS	Cepstral Means Subtraction
DCT	Discrete Cosine Transform
DET	Detection Error Trade-off
DFT	Discrete Fourier Transform
EER	Equal Error Rate
EM	Expectation Maximization
FFT	Fast Fourier Transform
FIR	Finite Impulse Response
FN	False Negative
FNR	False Negative Rate
FP	False Positive
FPR	False Positive Rate
GMM	Gaussian Mixture Model
GMM-UBM	Gaussian Mixture Model-Universal Background Model
GMM SVM	Gaussian Mixture Model Support Vector Machine
HMM	Hidden Markov Model
IDFT	Inverse Discrete Fourier Transform
LPC	Linear Prediction Coding
LPCC	Linear Predictive Cepstral Coefficients
MAP	Maximum a Posteriori
MFCC	Mel-Frequency Cepstrum Coefficients
NPC	Neural Predictive Coding
PLP	Perceptual Linear Prediction
PS	Power Spectrum
SVM	Support Vector Machine
UBM	Universal Background Model

This page intentionally left blank

1. Introduction

Recognition of body, face and voice has been utilized by humans for thousands of years to recognize other humans and objects. This is an ability that humans have from a young age to distinguish a mother, father, etc. [1][2]. There are a few occasions where one can be deceived by appearance, e.g. similar dressing, colors and so on [1]. Recognition in a larger scale can be difficult for human perception, and therefore it could help to find a way to automate this recognition, and the automated recognition of individuals based on biological characteristics is referred as biometrics [1].

A biometric system is an automated pattern recognition system [2], the operations of such a system is to capture biometric samples such as voice, face, fingerprints, etc. and compare them to already collected biometric reference samples that are already enrolled. The biometric system is then able to match the samples depending on different models and algorithms [1]. To further understand the biometric system functionality, it is best described through an illustration in figure 1.

The progress in digital electronics with the development of new algorithms and computing hardware has made it possible for people to utilize voice-based applications. The advantage of speech is to be able to express something, communicate and interact. A Speech signal consists of a sequence of sound waves, this sequence is formed through acoustical excitation of the vocal tract (comprising the throat, mouth and nasal cavity) when air is expelled from the lungs [3][4][5]. And with the development of voice-operated functions many high-tech products have been developed with biometric identification technology, e.g. Speech- and speaker recognition that utilizes speech signals to interact with TVs, smartphones, and home personal assistants, just to name a few. Controlling the lights in your home, setting alarms and locking the front door through your speech is possible with Amazon Echo and the recently launched Google Home in 2016 [6][7][8]. In most cases these devices are in a home environment and have control over things that need to be secure, and therefore a need for security technology has increased and has to be implemented. With security in focus then such technologies with Biometric recognition i.e. speaker recognition can be one solution [9][10].

To increase the recognition rate and security in the biometric system then a combination of biometric identification is possible e.g. combining face recognition and speaker recognition or speaker recognition with fingerprint recognition etc. The combined system is able to analyze and determine if the person is who he is claiming to be, if the system matches the person to the claimed identity, then person is the “client”. While if the system does not find a match in the system then that person will be regarded as an imposter [11]. Many applications today utilize voice recognition technology and depending on the nature of the application it is possible to see if the application is focused on **speech recognition** or **speaker recognition**. Speech recognition is the ability to recognize words from the human speech, while speaker recognition refers to the task of recognizing and identifying the person that is speaking based on the spoken utterance. Speaker recognition is then also divided into two parts, the first part is **speaker identification** and the second part is **speaker verification**. Speaker identification is to determine who is speaking based on a set of registered speakers, and speaker verification on the other hand is to analyze and then determine if the unknown voice is from the claimed speaker (true speaker¹) [7][9][12]. The speaker recognition system can also be modeled to be either text-dependent or text-independent. A text-dependent system is based on the same utterance “specific password” from the claimed speaker. While for a text-independent system there is no such constraint, the system will recognize the specific speaker based on other utterances “passwords” [11].

The speaker verification system is divided into two phases, enrollment and verification. The enrollment phase is to capture biometric samples and compare them to already collected biometric reference samples as mentioned above, and typically a feature extraction method is required such as Mel Frequency Cepstral Coefficients (MFCC) to extract the specific speakers features. The MFCC features can also be used for a text-independent system. After that is the verification phase which is the part to recognize the specific speaker based on features extracted, and statistical approaches such as Gaussian Mixture Model (GMM), Gaussian Mixture Model-Universal Background Model (GMM-UBM), Gaussian Mixture Model Support Vector Machine (GMM-SVM) model, Hidden Markov Model (HMM) and neural networks are preferred for matching [11]. In recent years, approaches for text-independent verification systems are GMM based (e.g. GMM-

¹ A true speaker is a person that has a speaker specific model in the system.

UBM, GMM-SVM) and has been almost exclusive [13]. HMM based approaches has not shown any advantages for speaker recognition systems that are text-independent compared to GMM based approaches [14].

“There is no concrete evidence that using the maximum amount of data would guarantee the best overall performance” [13]. Which in turn means that more training data would not necessarily give better performance because of the physical constraints of the humans that will limit the speech to only occupy a limited region in the feature space [13].

1.1 Research Motivation

The aim of this research is to investigate the performance of the true speaker data collection in a GMM-UBM based speaker verification system, the reason for this research is to conclude on how much data is necessary for the system, as Taufiq, H. et al. [13] said in his research that there is no concrete evidence that says that too much data is “always better”. This research will therefore give a contribution to how much data is necessary for a secure system where the False Positive is as close to zero as possible, how will the amount of training data affect the False Negative (FN). The main goal of this study is to provide the researchers with knowledge of how much training data is needed for a speaker verification system, this will save the researchers a lot of time and funds. To further investigate the area of security in a speaker verification system then two methods are compared (GMM and GMM-UBM) to experiment on which is more secure depending on the amount of training data available. This will help the researchers to choose the right method at an early stage depending on the system criterias they set up.

	GMM compared to GMM-UBM with true speaker model (data less than 15 sec)	GMM-UBM imposter model (amount of data needed)	GMM compared to GMM-UBM with true speaker model (data between 45 sec to 1 hour)	GMM-UBM true speaker model (amount of data between 45 sec to 1 hour)
This Research			X	X
Kumari, T. J., Jayanna H. S.[28]	X			
Pelecanos, J. et al. [32]		X		

Table 1: The main contribution of this research compared to related work

The papers in chapter 3 are related to this thesis in many ways, e.g. how to extract features from an utterance or how the enrollment phase is set up and or algorithms for the verification phase, but considering the research question than two papers are highly related to this thesis and are therefore presented in Table 1.

1.2 Research Question

The aim of this research is to investigate the performance of the true speaker data collection in a biometric speaker verification system, and therefore propose this research questions below.

Research Question One: How does the GMM-UBM speaker verification system perform based on different amount of data collected for the true speaker model?

Research Question Two: How does the GMM speaker verification system perform compared to the GMM-UBM, based on different amount of data collected for the true speaker model?

1.3 Limitation

The limitations and approaches for handling threats to validity will be brought up in this section. The verification system is based on the program Matlab in a personal computer. In the enrollment phase the use of MFCC for feature extraction is used with GMM-UBM- and GMM models for the verification phase of the system. The imposter model contains two databases, one database contains approximately 5.4 hours of data and the other one around 1 hour of data which is in total around 6.4 hours of data that is used for training the imposter model. The imposter data consists of 70% male speakers and 30% female

speakers. The evaluation of the results in chapter 5.5 are based on 9 male- and 1 female speaker. The training of the imposter data is recorded with two different microphones for avoiding any biased results. One of these microphones was used when collecting true speaker data and evaluation data. Two true speakers (male) will be evaluated in sixteen models (eight for GMM classifier and eight for GMM-UBM classifier) each, and the amount of training data consist of 45 seconds up to 1 hour without any use of silence removal.

2. Theoretical Background

The aim of this chapter is to review the areas that are important to understand in order to follow the chapters ahead in this thesis. Each subsection below should give a sufficient understanding to each term.

2.1 Biometrics System

Biometric is the physical characteristics of a human being (e.g. voice, fingerprinting, face). A biometric system is a pattern recognition system that acquires biometric features from human beings, that are compared with the features stored in a database.

There is a variety of biometric characteristics systems used (e.g. DNA, face, fingerprint, signature, voice, iris, etc.) each has its strengths and weaknesses [2][15].

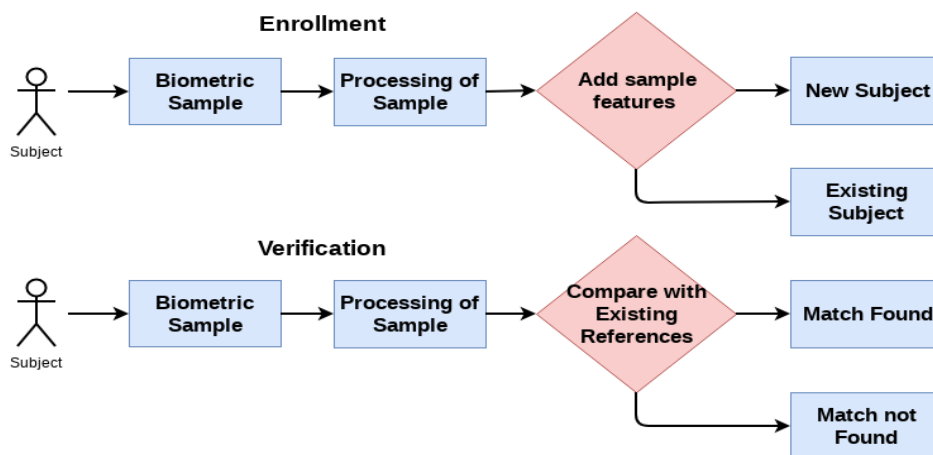


Figure 1: The operations of a typical biometric system [1]

The biometric system is divided into enrollment and verification. In the enrollment phase the system decides if the subject is existing in the system “database” and if not then the subject should be added for future reference. While in the verification phase the system compares the subject with existing references and returns a match if found [1].

2.1.1 Verification System

In the verification system, a person claims to be a certain user whose biometrics features are saved in a database. This is done by providing ID (can be PIN number, name, RFID tag etc.) then the model is retrieved from the database to be compared, if the claimed person is the intended user or an imposter. In this way, the system is validating the person's identity e.g. person one is the true speaker [2][16].

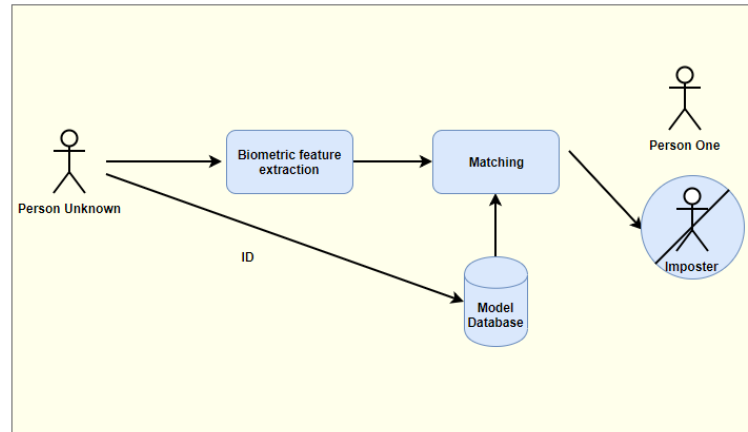


Figure 2: The verification process

2.1.2 Identification System

An Identification system operates to recognize an individual by searching the “database” of all the models of the users, and to find a match. This is the main difference between verification (one-to-one comparison) and identification (one-to-many comparison) [2][16].

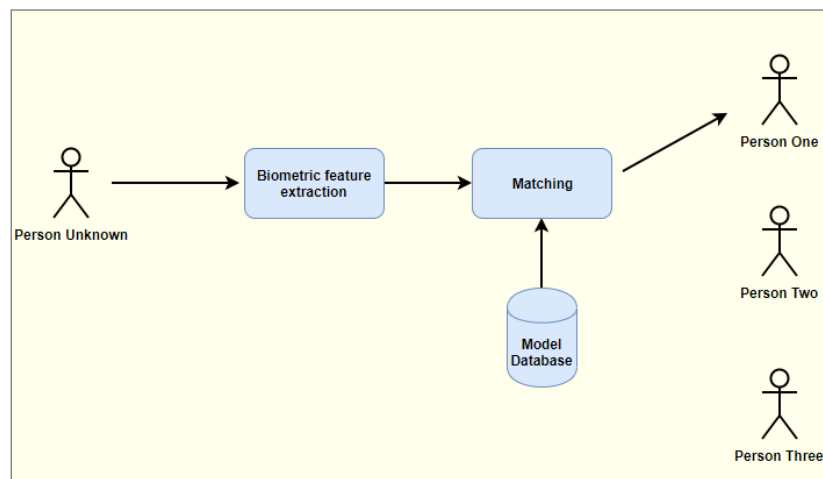


Figure 3: Identification phase with a set of speakers

2.1.3 Speaker Verification System

Speaker verification system can be divided into two modalities, text-independent and text-dependent. A text-independent identification system relies on the characteristics of the vocal tract of the speaker. This is a more complex modality which can offer better protection against imposters. A challenge of text-independent verification systems is that the features are more sensitive to background noise. While for text-dependent verification systems, they are dependent on a fixed phrase that is predetermined [2][16].

2.1.4 Performance Metrics for Biometric Systems

One popular performance metric in biometric systems is the Equal Error Rate (EER). Biometric systems have a trade off between False Negative Rate (FNR) and False Positive Rate (FPR) and the EER is a measure of the error rate of the system when the threshold is varied so that the number of FNR is equal to the number of FPR [16].

The Detection Error Trade-off (DET) curve was introduced by National Institute of Standard and Technology in 1997 and has been favored since then. The DET curve plots the percentage FNR on Y-axis versus percentage FPR on X-axis and the DET curve uses a logarithmic scale for plotting the two rates [2][16].

		True Condition	
		Condition Positive	Condition Negative
Predicted condition	Predicted Condition Positive	True Positive TP	False Positive FP
	Predicted condition negative	False Negative FN	True Negative TN

Figure 4: EER metric structure [40]

To calculate the FNR and FPR then these equations are as followed.

$$FNR = FN / (Condition\ Positive) = FN / (FN + TP) \quad (1)$$

$$FPR = FP / (Condition\ Negative) = FP / (FP + TN) \quad (2)$$

2.2 Digital Signal Processing

Signals are patterns of variations that represent information. There are all kinds of signals such as speech signals, audio signals, video or image signals, radar signals, just to name a few [17] [18].

Many signals originate as continuous-time signals, and speech signals are one of these. It can sometimes be desirable to obtain the discrete-time representation of the signal, and one way to do this is through sampling equally spaced points in time. The result will be a discrete time representation of the signal that can be processed digitally [17]. Digital Signal Processing (DSP) refers to a set of algorithms that are used to process digital signals, the usage for some of these algorithms is to improve the signal by using techniques such as Discrete Fourier Transform (DFT), Finite Impulse Response (FIR). Other algorithms that are used with DFT and FIR are Windowing [19].

2.2.1 Frame-Blocking and Frame-Overlapping

Frame-blocking is to split the signal into frame blocks, the frames are usually between 10 to 30 milliseconds. This is necessary in order to capture the local spectral properties that are assumed to be stationary. Then to smoothen the transition from one frame to the other, an overlapping technique is used [20].

2.2.2 Hamming Window

A hamming window is used on each frame of the signal. This is needed to remove or reduce any signal discontinuities at the start of the frame and at the end of the frame when doing Fourier analysis. For MFCC the most common window used is Hamming window, followed in equation 3 [2][16][21].

$$F_k = 0.53836 - 0.46164 \cos\left(\frac{2\pi n}{N-1}\right) \quad n = 0 \dots N - 1 \quad (3)$$

2.2.3 Discrete Fourier Transform

DFT is to transform a signal from time domain to frequency domain. This is done by taking a sequence of sampled data and compute the frequency content in it, e.g. extract the frequencies that the speech signal is composed of. This represent the frequency domain of the signal opposed to the time domain. This is a powerful tool in DSP applications, this allows an examination of the frequencies in any given signal [17][22]. The Fast Fourier Transform (FFT) is an optimized algorithm of the DFT [17][22][23].

2.2.4 Spectrum

The spectrum of a speech signal is referring to the frequencies that are extracted with a Fourier transformation on a single frame [20], this is a powerful tool which provides the spectral content of the speech signal. And with the Inverse Discrete Fourier Transform (IDFT) it is possible to go back to the time domain. Simply put, a plot of the time domain will reveal the changes that are occurring over time, while for the frequency domain it specifies how much of the signal is in a given frequency band over a range of frequencies [22][24].

2.2.5 Cepstrum

The cepstrum can be considered to be a spectrum of a spectrum, mathematically it is the IDFT of a logarithm of the magnitude of the DFT of the signal. Cepstrum is more preferred in this case instead of the regular spectrum because it has the ability to separate the spectral envelope and the excitation signal from the spectrum. The process is that when taking the IDFT of the spectrum a lower frequency “spectral envelope” and a higher frequency “excitation signal” will be extracted. The main reason for this is to “filter out” the higher frequencies and attain the vector of cepstrum coefficients [24].

2.2.6 Mel-Frequency Cepstrum Coefficients

The MFCC method is widely used in Speaker verification systems. The aim of a feature extraction technique is to characterize an individual’s vocal tract information, and the MFCC is popular because it is based on the known variation of the human ear’s critical frequency bandwidth. In this paper the MFCC, Delta and Delta-Delta is built with the following blocks seen in figure 5 [25]

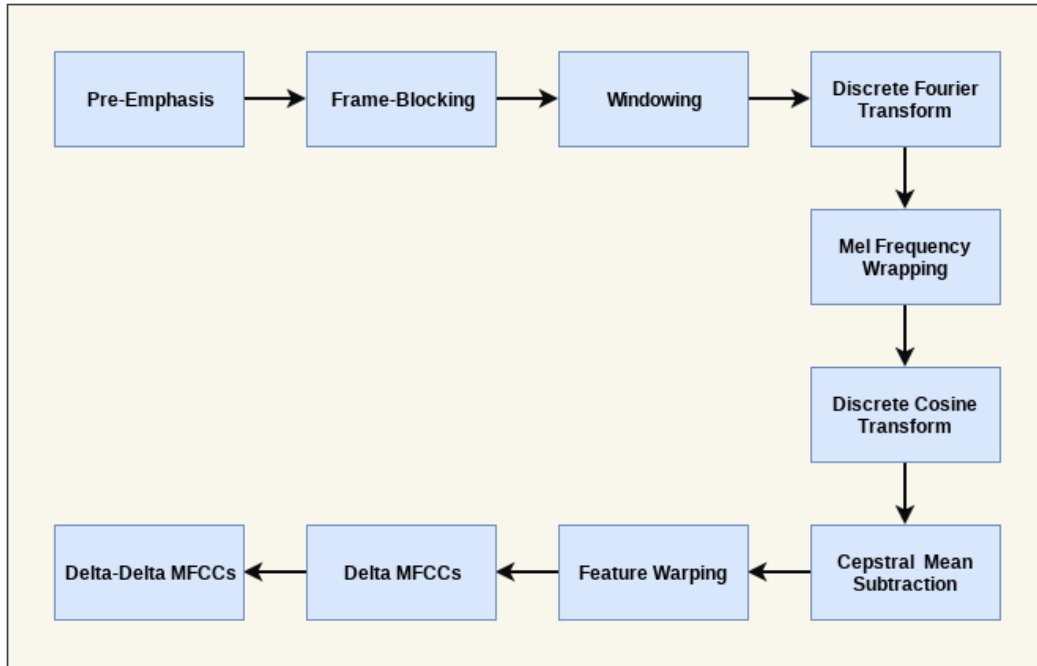


Figure 5: Block diagram of MFCC with Delta and Delta-Delta

2.3 Speech Signals

Speech is a tool humans use to communicate, and the speech signal is generated through the mouth when air is moving from the lungs to the throat and through the vocal tract. This tool is one way of expressing languages. And speech can be characterized in terms of signals carrying a message, which is the acoustic waveform [22]. Speech signals are based on the physical shape of the person, mouth, vocal tracts, age, medical conditions and emotional state. These conditions can change the speech signals over time [2].

2.4 Voice Modeling

There are two important parts in speaker recognition, the first part (front-end) which handles feature extraction that is to extract the features from a speech signal, and the second part (back-end) is the modeling and verification phase, this can be seen in figure 6. The most common modeling techniques used are HMM and GMM. The main difference to consider for the speaker verification system is that HMM is text-dependent and GMM is text-independent [24][26].

2.4.1 Gaussian Mixture Model

The GMM is a powerful tool for modeling text-independent system (has also been used in text-dependent systems). GMM is used in many applications that are involved with statistical data analysis, pattern recognition, computer vision, voice and image processing, and machine learning. The GMM is able to handle difficulties with data analysis and clustering with the finite mixture of gaussian densities. GMM handles clustering which is an unsupervised classification of data put into groups (clusters). This makes the GMM a useful probabilistic model for clustering of data, and current state-of-the-art text-independent GMM systems obtain the background model parameters by using the Expectation Maximization (EM) algorithm [21]. Example of application areas that utilize GMM is traffic surveillance, foreground segmentation, image coding, tracking and classifying activities, to name a few [24][27]. The mathematical form of GMM is seen in equation 4 below and is defined as a parametric probability density function and is represented as a weighted sum of M gaussian components [27][27][28]. The advantage of the GMM method compared to the GMM-UBM is that it requires less training data for the imposter model, while a disadvantage is that it requires more training data for the true speaker model [28][29].

The GMM method is a likelihood function with a mixture of M Gaussians illustrated in the equations below [21].

$$p(x|\lambda) = \sum_{i=1}^M w_i p_i(x) \quad (4)$$

$p(x|\lambda)$ is the frame based likelihood function, λ is the hypothesis or likelihood function, x is a set of features (MFCCs), and $p_i(x)$ is the individual Gaussian density function.

The parameters for the model are the mixture weight w_i , the mean vector μ_i , and covariance matrix Σ_i which in short can be written as $\lambda = (w_i, \mu_i, \Sigma_i)$ where $i = [1...M]$.

$$p_i(x) = \frac{1}{(2\pi)^{n/2} |\Sigma_i|^{1/2}} \times \exp \left\{ -\frac{1}{2} (x - \mu_i)' \Sigma_i^{-1} (x - \mu_i) \right\} \quad (5)$$

The next step after the model has been trained is to evaluate the log-likelihood of the model **with** a test set of feature vectors MFCCs, X [21]:

$$\log p(X|\lambda) = \sum_{t=1}^T \log p(x_t|\lambda) \quad (6)$$

2.4.2 Universal Background Model

The UBM is basically a text-independent GMM trained with the features extracted from a set of speakers. This is identified as a large combination of speakers that can be regarded as “imposters”. This is a suitable strategy to utilize for the GMM-UBM, the difference is that instead of making a model for each individual speaker (GMM), instead the speaker model is derived from the UBM with the Maximum a Posteriori (MAP) estimate algorithm, the UBM consist of many speakers and in this way it is utilizing all speaker data to enhance the identification rate. The reason to use UBM is to overcome difficulties such as insufficient training data or “unknown data” [29]. As mentioned in section 2.5.3, the UBM requires less data for the true speaker modeling and it gives lower EER with limited true speaker data [21][28][29].

The EM algorithm that is mentioned in section 2.5.1 is used to iteratively improve the model parameters estimates and to maximize the likelihood in around 5-10 iterations. The GMM-UBM method is used for speaker model training where the MAP estimation technique is used to adapt the background model to each enrolled speaker. In this way the speaker model is affiliated with the background model. The GMM-UBM method utilizes the trained parameters $\{w_i, \mu_i, \Sigma_i\}$ from the EM algorithm. And finally the GMM-UBM can be applied to the likelihood-ratio score.

$$\Lambda(X) = \log p(X|\lambda_{hyp}) - \log p(X|\lambda_{\overline{hyp}}) \quad (7)$$

The method $p(X|\lambda_{hyp})$ is the true speaker model and $p(X|\lambda_{\overline{hyp}})$ is the imposter model [21].

An illustration of the GMM-UBM can be seen in figure 6.

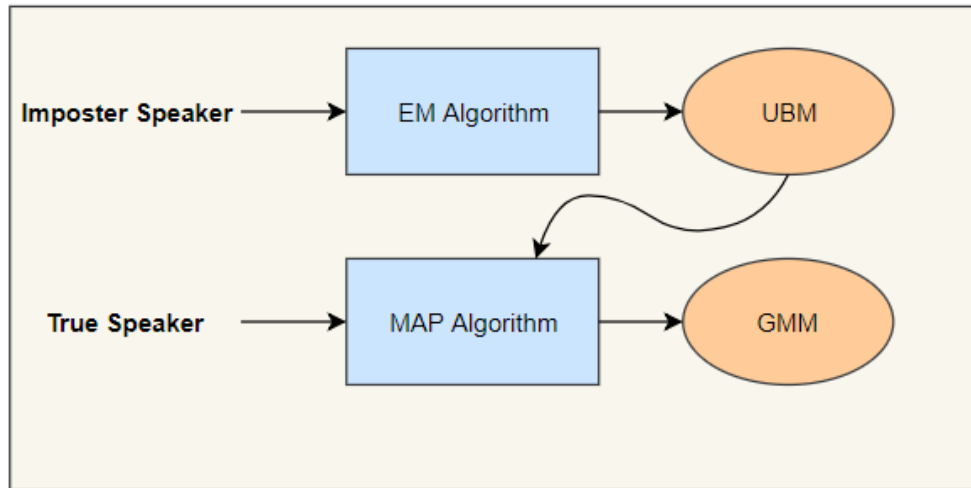


Figure 6: The process of the GMM-UBM [21].

3. Related Work

In this section, you will find relevant information contributed from previous work that is closely related to this thesis.

3.1 Hossan. M.A., et al. - A Novel Approach for MFCC Feature Extraction

This paper presents the MFCC feature extraction method in a detailed step by step explanation with algorithms that are easy to follow and figures. In the paper they explain the reason for the use of MFCC for speaker recognition systems, which is that the MFCC is based on known variations of the critical frequency bandwidth of a human ear, and they also mention other methods such as Linear Prediction Coding (LPC), Linear Predictive Cepstral Coefficients (LPCC), Perceptual Linear Prediction (PLP) and Neural Predictive Coding (NPC). To extract the features, they de-correlate the output log energies of a triangular filter bank which is linearly spaced on the Mel frequency scale and to de-correlate the speech they use the Discrete Cosine Transform (DCT) implementation. In this paper the proposed method is to extract the MFCC features they say that 18 coefficients should be enough to represent the 20 filterbank energies. To further obtain more detailed speech features they use a derivation on the MFCC vectors (DMFCC) and the second order derivatives (DDMFCC). For the classification and verification stage they use GMM with EM, and they explain that these are widely used algorithms for speech and even pattern recognition. And to further enhance the outcome they used UBM-GMM and MAP-GMM. The results that are presented in this paper show that the use of GMM-EM and DCT based MFCC gave an accuracy of 96.72% [25].

The paper gives a detailed and thorough analysis for each block of the MFCC method which is important to understand in order to answer the following research question(s) set in chapter 1.2. Therefore, it is important to evaluate if 18 coefficients is enough to represent e.g. 20 filterbank energies, and further evaluate the accuracy improvements of the derivatives.

3.2 Fang, Z., et al. - Comparison of Different Implementations of MFCC

In this paper the focus is on comparing the different available MFCC implementations to evaluate the performance. They make several comparisons and the experiments are on the number of filters, the shape of the filters, the spacing of the filters and the way the Power Spectrum (PS) is warped. The paper first introduces the MFCC blocks and explains the gravity of the extraction phase and how it can significantly affect the recognition phase. An important point that is brought up in this paper is that the first coefficient is ignored and they prove why this is so with experiments. Further they inform why the use of first order and second order time derivatives are important and many experiments have shown that this can improve the performance of the system. They illustrate the results with tables that are easy to understand along with an analysis and a recommendation based on what system that is intended to be used [30].

In this paper the use of MFCC is used for feature extraction, therefore this paper has given many good points that are important to analyze such as the number of filters, the shape of the filters, the spacing of the filters, just to name a few.

3.3 Chao-Yu, L., et al. - User Identification Design by Fusion of Face Recognition and Speaker Recognition

This paper explains how security is essential in many applications and systems, and to overcome this hurdle they propose a fusion of recognition systems that are based on both face- and speaker classifiers to improve the recognition rate in 15 environmental situations. The recognition system they design is divided into two phases, the registration- and the recognition phase. The registration phase is commonly used in biometrics systems and is to register the speaker specific data. And the recognition phase handles the recognition of the speaker if it is an imposter or not. In the recognition phase they utilize the GMM-UBM modeling for text-independent speaker recognition. The idea of having a fusion of two classifiers is to receive the prediction results from the classifications based on the threshold for the two classifiers determine the results or recognition. For the system they utilized a database with 30 different faces and approximately 160 words from different speakers. For the training of the system they used 20 samples of face and 10 samples for testing, and 144 words for training and

10 sentences with 3-4 words for testing. The results that they acquired shows that in 15 different situations the experiment gave a recognition result that outperform any single classifier in each of these 15 situations [11].

This paper gives a broad insight for security and therefore they propose a fusion of classification techniques, but also a detailed description of the GMM-UBM modeling that is essential in text-independent applications such as speaker verification and recognition. They had a recognition rate of 93.3% which can be discussed in the further chapters, since they had only around 160 words which can be considered a bit too low for the UBM-GMM system. Therefore, this paper will help us answer the research questions that are set in chapter 1.2.

3.4 Mohamed, K.O., et al. - Training Universal Background Models for Speaker Recognition

This paper explains the concept of GMM, that it is used for modeling in speaker recognition but later brings up the challenges that can be found such as the need for a large amount of data to train the model for the maximum likelihood criterion. Therefore, they present UBM and three alternative criterias for the training of UBM. The speaker recognition system is designed with the use of GMM vectors and the MAP algorithm, and the UBM is trained with GMM using the EM algorithm with a large amount of data such as a database with different speakers, the speakers are equally divided in gender for best results. The database used contains a collection of 13770 utterances, where 6038 utterances are from male speakers and 7732 utterances of female speakers. As many other biometric systems, they have both front-end and back-end. The front-end is to extract the speaker specific features, for this they use the MFCC method. The feature extraction consists of 36 dimensional features extracted from 12 cepstral coefficients along with their corresponding first order derivative “delta” and second order derivative “delta-delta”. They further choose to set the filterbank with 24 filters over a frequency range of 125-3800 Hz. The signal was windowed with 32ms of length along with a 10ms frame overlap. A normalization technique feature warping was then applied to the MFCCs to reduce linear channel and slowly varying additive noise effects. For the back-end they trained the GMM model with a dimension of 36864 for each utterance, and this was achieved by using a UBM of 1024 gaussian components with the MAP

adaptation algorithm. The results that are presented shows that the phonetically inspired UBM (PIUBM) system outperforms the other two systems that are presented in this paper [31].

This paper is highly relevant for the research questions that we are going to answer in the following chapters, they give a detailed explanation of both the front-end with the MFCC method and its parameters along with the back-end with the UBM-GMM method.

3.5 Taufiq, H., et al. - A Study on Universal Background Model Training in Speaker Verification

This paper focuses on the UBM training for a speaker verification system, they shortly present the GMM method that has received a lot of attention for its ability to be text-independent in speaker- recognition and verification systems. The aim of their paper is to analyze the performance of the speaker verification system based on the UBM data. The idea is to alter the UBM data and analyze the resulting EER of the system. For this they are going to experiment with variations in the amount of data, the structure of the feature frames in the front-end, and a variation in the number of speakers. Like many others they structure their system with the GMM based speaker recognition methods with the MAP adaptation algorithm of the UBM (GMM-UBM). They further explain that the use of UBM is a crucial factor for their research, the UBM is a large data set trained GMM that is representing the speaker independent distribution of the speech features for all speakers. They further explore another GMM based system which is the **GMM-Support Vector Machine** (SVM) modeling. They further narrow their research towards the amount of speakers of the UBM, saying that it is today a common issue to use as many speakers as possible and lack consideration towards performance tradeoff. An issue to address is the construction of the UBM with regard to performance. They begin by dividing the UBM training process into two categories, the algorithm parameters and the data parameters. The UBM algorithm parameters consist of number mixtures, method of training, number of iterations, method of initialization, etc. The second part is the training data parameters which is relevant to the research question set in section 1.2 in this paper, and is about the amount of data, number of speakers in the data, amount of data per speaker, method of selecting speakers, ways of using the

feature vectors, etc. They argue that there is no concrete evidence for the assumption in the UBM training that many tend to interpret as “the more data used the better the system performance will be”. Their results indicate that 2.7 seconds utterance should be sufficient for the UBM with 1.5 hours of training compared to 260 hours [13].

This paper brings up many important points to consider when answering the research question in section 1.2. Especially the part where they have different experiments to analyze the performance of the UBM based on a variety of speakers and the amount of data for each speaker. Other key points that they mention is the feature extraction (front-end), Modeling (GMM-UBM) and their corresponding mathematical algorithms.

3.6 Pelecanos, J., et al. - Feature Warping for Robust Speaker Verification

Normalization is an important part in speaker verification systems, this paper presents different normalization techniques that is intended to improve the performance of the system. These techniques are computed on the feature extraction phase of the system, to normalize and compensate for the effects of environmental noise that can occur due to different training and testing environments. The techniques that are compared in this paper are feature warping, Cepstral Mean Subtraction (CMS), modulation spectrum processing, and short-term windowed CMS and variance normalization. Different normalization techniques have been experimented with such as CMS, which was a promising approach that was intended to remove linear channel effects. However, under additive noise conditions they noticed that the feature estimates would degrade significantly. Therefore, this paper's aim is to investigate and compare the feature warping normalization, which is intended to construct a robust representation of each cepstral feature distribution. For the enrollment phase like many others they have used the MFCC method to extract the cepstral features. And for the verification phase they use the GMM-UBM method for modeling and verification. The system was based on 230 male and 309 female target speakers where each speaker had a recording of 2 minutes of speech for training. A total of 1448 male and 1972 female test segments with a minute of recording was used. The results show that the feature warping normalization was the better performing technique compared to the other algorithms proposed in this paper [32].

As mentioned in speaker verification systems a normalization of the MFCCs is an important part for improving the performance of the system. Therefore, this paper gives a good insight in what each normalization technique is intended for and what to think about before implementing them in the enrollment phase.

3.7 Kumari, T. J., et al. - Comparison of LPCC and MFCC features and GMM and GMM-UBM modeling for limited data speaker verification

In this paper the focus is on comparing the EER for two different methods for feature extraction and comparing the EER for two different modeling for classification for speaker verification system. The comparison is made with limited true speaker data (less than 15 seconds). The feature extraction methods used in this paper is MFCC and LPCC, and the method for the classification methods are GMM and GMM-UBM. The methodology used in this paper is a controlled experiment, the variables that are changed for each experiment are the feature extraction (LPCC vs MFCC), the amount of features used (13 LPCC/MFCC vs 39 DDLPC/DDMFCC), and classifiers (GMM vs GMM-UBM). The database used for this experiment is the NIST2003 database, the models created with 3,4,5,6,9 and 12 seconds duration. The results show that the use of 39 features was superior for both MFCC and LPCC. LPCC performed better than MFCC in all cases. The GMM-UBM gives less EER compared to GMM with limited data [28].

This paper gives good insight on GMM and GMM-UBM, the advantages and disadvantages, and the use of 39 coefficients for feature extraction for better data. Their research is tightly coupled to ours and they focus on limited data such as 3-12 seconds which will be interesting to discuss further in the chapters 6 and 7.

4. Method

The research process of this thesis is based on Nunamakers et al. [33] suggestion on system development research methodology, which is a developmental, engineering and formulative kind of research. The framework that is proposed by Nunamaker et al. consists of six stages, *Construct a Conceptual Framework*, *Develop a System Architecture*, *Analyze and Design the System*, *Build the (prototype) System*, *Observe and Evaluate the System*.

The system development research methodology framework shown in Figure 7 is chosen mainly because of its exceptional structure for development of systems. The framework is a combination of both iterative design and an incremental build model. This ensures that the system will be developed through incrementation of portions and iteratively to ensure good quality.

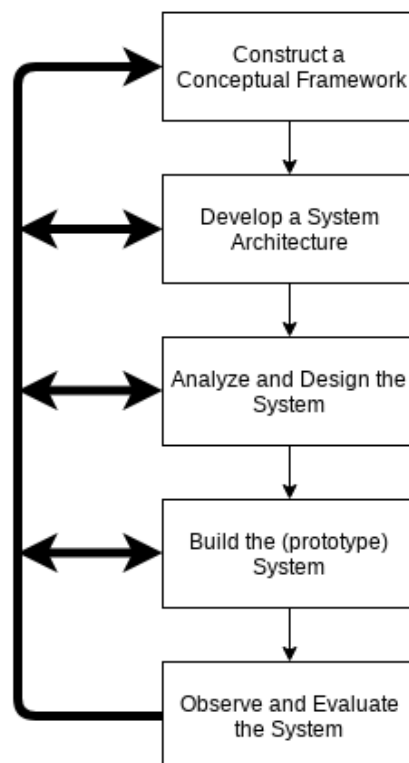


Figure 7: Systems development Research Methodology [33]

4.1 Construct a Conceptual Framework

The first stage of the framework is about information gathering and problem formulation, at this stage, many literature reviews are researched to gather as much information on the problem domain as possible and to formulate the research questions set in section 1.2.

To extract the most relevant information for the research questions then a so-called “drilling in” will be conducted to find as much rich data as possible about voice biometrics and the applications that are related to voice biometrics, the front-end (feature extraction) and the back-end (modeling and verification) methods of these applications. The main objective is to gain knowledge on the methods used for feature extraction and the speaker verification models that are related to voice biometric systems. The next phase will be to construct the conceptual framework by abstracting out the raw data and gather the manageable, understandable and meaningful data.

4.2 Develop a System Architecture

The second stage of the framework is to develop a unique system architecture to get a better understanding of the system components and the relationship between them. The system components are divided into front-end and back-end. The front-end of the system takes care of speaker specific speech signals. The back-end handles speaker models, storage, and matching of speakers.

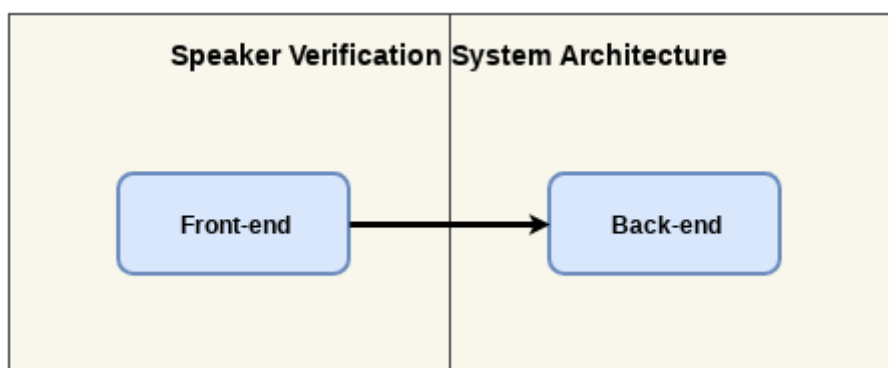


Figure 8: A common system architecture for speaker verification

The system architecture in figure 8 is a common speaker verification system architecture. At this stage it is important to consider all the different methods that exist and are being used in the front-end, and the same goes for the back-end of the system. Therefore, a more detailed research of these two blocks will be conducted and illustrated in section 4.3.

4.3 Analyze and Design the System

The third stage of the framework is regarded as the most important stage for the development process, at this stage the knowledge gained on the domain is put into practice and the design will act as a blueprint for the prototype system. By further exploring the speaker verification system domain, it should be clear on which methods, models, etc. are involved for the speaker verification system.

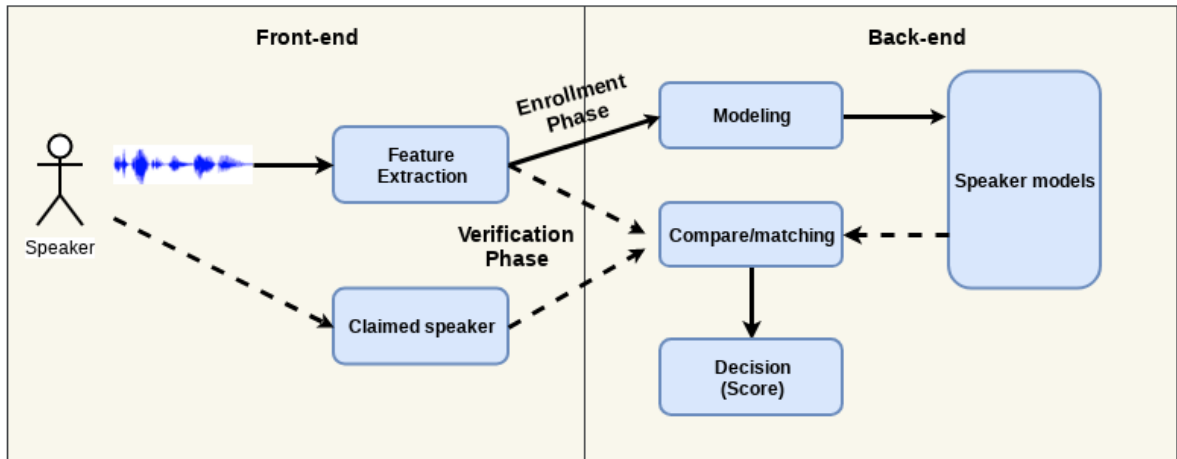


Figure 9: System Architecture and components for a speaker verification system [34]

The front-end handles the feature extraction of the claimed speaker. And the back-end is further divided into enrollment phase and verification phase. The enrollment phase consists of modeling which creates and later stores the speakers model. The second phase is the verification phase, here the system will compare and match the claimed speaker to an existing reference model, and lastly decide if the claimed speaker is an imposter or not.

4.4 Build the (prototype) System

At this stage the system architecture is ready to be built in a system. The system should be able to meet the expectations that are set and answer the research questions defined in section 1.2. The system is built iteratively for best results. The advantage of building a system at an early stage makes it possible to see the advantages and disadvantages of the system design, and this in turn will give the researchers new insights to evaluate, and if necessary be able to re-design the system architecture.

The process and components involved for the system is described in section 5, where a more thorough and detailed explanation on each part is presented along with the results of the system.

4.5 Observe and Evaluate the System

The last stage is to evaluate the system, the researchers are able to observe and evaluate the system performance and usability. The data that is presented in subsection 5.5, is then interpreted (subsection 5.6) and discussed (chapter 6) in order to decide if the system is satisfactory to answer the research questions set in section 1.2. The observations will be discussed in more detail in section 6, where an evaluation will be presented of the results.

5. Results

The speaker verification system results are presented in this section. The results are computed with the MFCC method for the front-end (feature extraction) in Matlab and the back-end (verification and modeling) are based on the GMM and GMM-UBM methods also in Matlab.

5.1 The System

The functionality of the speaker verification system is to determine whether a certain speaker is the claimed person by comparing a speech utterance of the speaker with an already existing, pre-recorded reference speech model for the claimed speaker. This system is intended to be used in applications where identification-verification is needed, e.g. alternative to pin-code for a mobile phone, access to door stations, signature for bank errands, etc.

The speaker verification system will register a person's utterance and create a reference model for that specific speaker. When the person wants to gain access to the system then a verification of the person needs to be done. The first step for the person is to specify who he is claiming to be (a number from a set of speakers) and then make an utterance similar (text-independent) to the reference utterance that is pre-created.

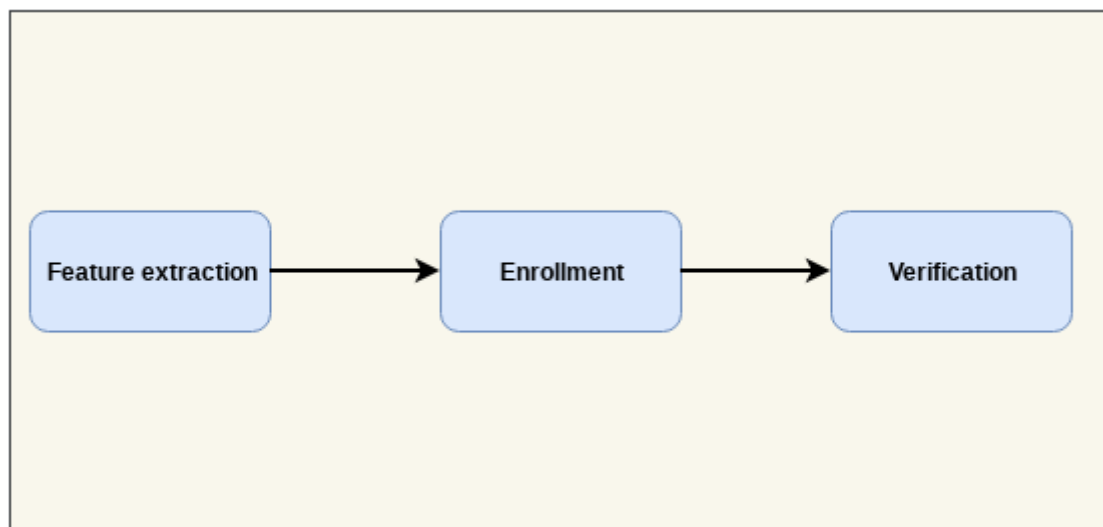


Figure 10: The three main components for a speaker verification system

As mentioned in section 4.3 there are three phases, the feature extraction phase (section 5.2), the enrollment phase (section 5.3), and verification phase (section 5.4). And to be able to answer the research questions, then the system needs to be done systematically/top down, this can be seen in figure 10. In the next subsections a more detailed description of the system will be presented.

5.2 The Feature Extraction Phase

To get good results and accuracy in a speaker verification system, it is important to have a feature extraction method that extracts the individual's vocal tract information along with feature normalization techniques in order to extract the relevant information.

5.2.1 Mel-Frequency Cepstrum Coefficients

When developing a speaker verification and identification system a feature extraction method is essential to capture the speaker specific information. Therefore, the chosen feature extraction method is MFCC which is a data set that represents basically the melodic cepstral acoustic vector of the speaker's speech. The chosen method is used in many related works such as “Comparison of Different Implementations of MFCC [30]”, “A Novel Approach for MFCC Feature Extraction [25]”, “Comparison of MFCC and LPCC for a fixed phrase speaker verification system, time complexity and failure analysis. [41]”, just to name a few.

The acoustic vector in this project is the feature vector and consists of 13 MFCCs. When used in real time the application with MFCC computes approximately 500% faster than LPCC [41], and is more robust to noise than LPCC[42].

5.2.1.1 Sampling

The sampling frequency is 44100 Hz with a bit/sample rate of 64 where the data is normalized to a range between -1 to 1, this is the default value set from Matlab.

5.2.1.2 Pre-Emphasis

About 80% of the power contained in a signal is in the lower frequencies 0-1000 Hz, while at higher frequencies the rate of power is dropped at around -12dB/octave [16]. When a signal passes through the pre-emphasis filter it will lower the power in the lower

frequencies to have a more even power distribution. The pre-emphasis is a first order high-pass FIR filter and the coefficient is 0.95 [16][35].

Figures 11 below shows before and after the pre-emphasis filter. A and B shows the speech signal in the time domain, while in C and D the Power Spectrum (PS) is shown in the frequency domain with the sample frequency 44100 Hz.

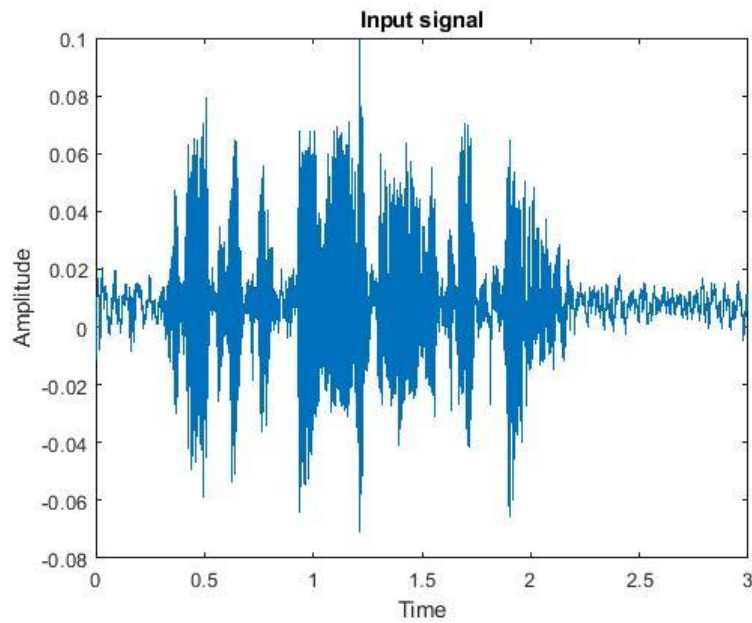


Figure 11A: Original signal

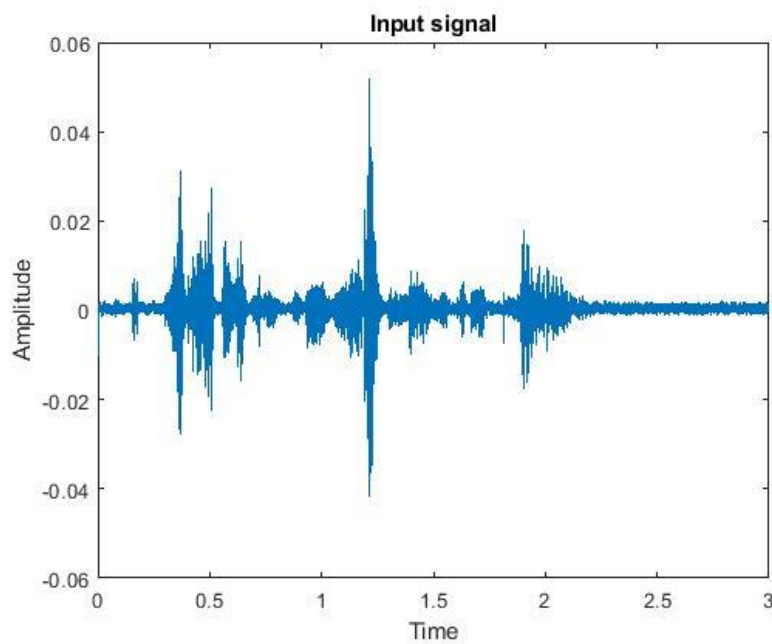


Figure 11B: Pre-emphasis signal

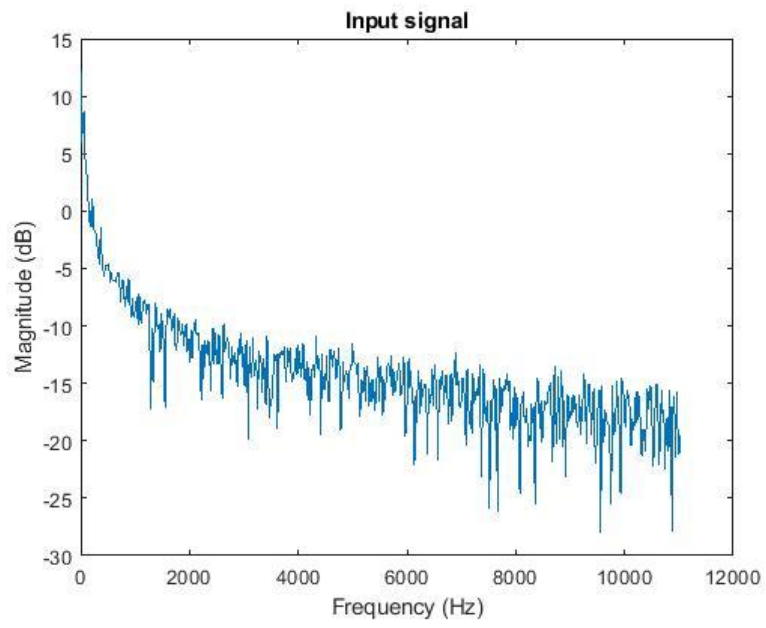


Figure 11C: Original signal (PS)

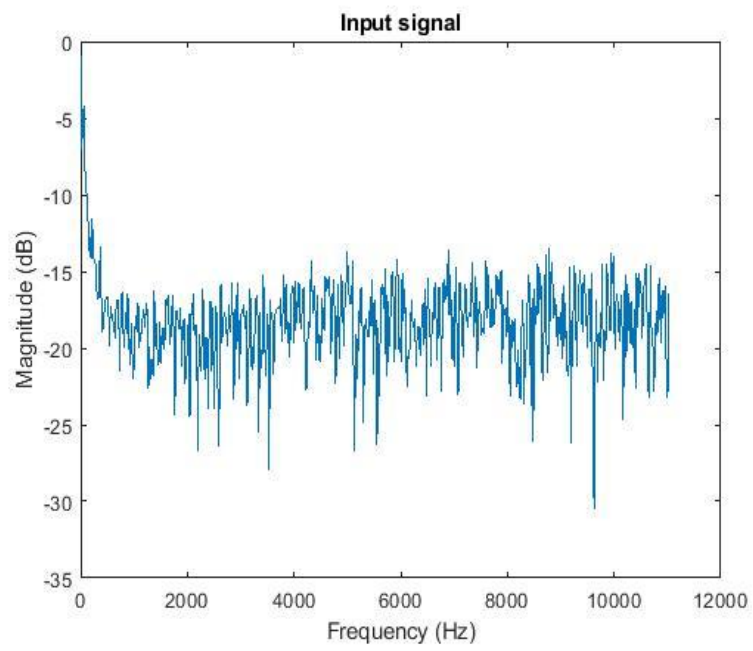


Figure 11D: Pre-emphasis signal (PS)

5.2.1.3 Frame Blocking

The next step after pre-emphasis is to split the signal into frame blocks, the frames are usually between 10 to 30 milliseconds. This step is important because in the next steps the use of Fourier transform is used, and it is important to capture the local spectral properties.

To further smooth the transition from one frame to the next frame, a frame overlapping technique is used, the purpose is to approximately center the sequence of the signal. Frame overlapping for MFCC is commonly between 30% to 70%. An overlapping of 50% is used [16][35][36][37], with 20 milliseconds frame size. The frame size is chosen because of the sample frequency which is 44100 Hz, this gives us even samples to overlap by 50%, illustrated in equation 8.

$$\frac{f_s \times \text{frame time}}{2} = \text{amount of overlapt samples} = 441 \quad (8)$$

5.2.1.4 Hamming Window

In this step a window is used on each frame of the signal. This is needed to remove or reduce any signal discontinuities at the start of the frame and the end of the frame when doing Fourier analysis. For MFCC the most common window used is Hamming window, followed in equation 9 [16][37].

$$W[n] = 0.54 - 0.46\cos(2\frac{\pi n}{N-1}) \quad (9)$$

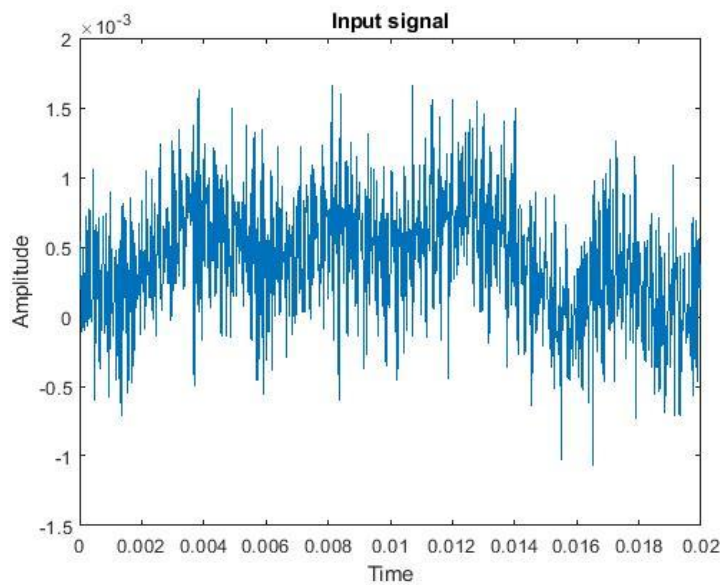


Figure 12A: A frame from the speech signal

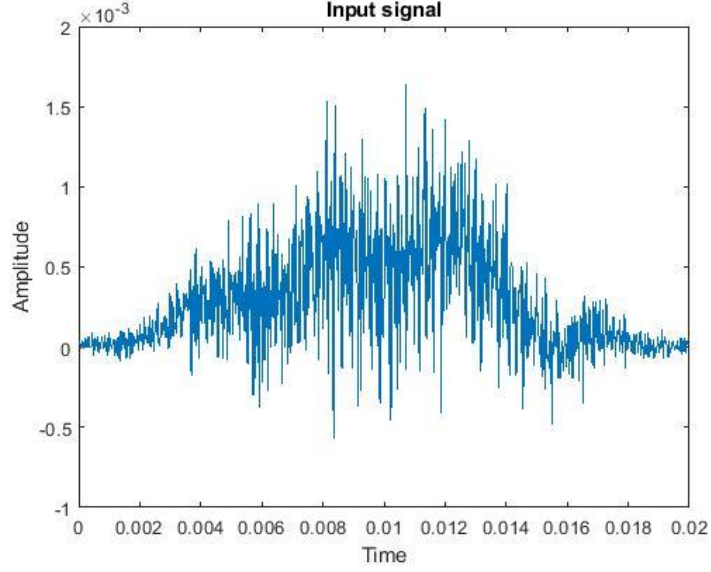


Figure 12B: Hamming window on the frame

In figures 12 an illustration is shown on how the hamming window technique works. The plotted frame in A and B is from the same speech signal. One can see in figure 12B that the amplitude at the end and the beginning of the signal is reduced compared to figure 12A.

5.2.1.5 Discrete Fourier Transform

The next step is to take the DFT of each frame with the use of the FFT algorithm seen in equation 10. This is done to extract the spectral information, and to know how much power is contained in the different frequencies [16][17][23][30]. To get the PS then taking the squared amplitude of the DFT is required [5]. The MFCCs are derived from the DFT of the frames along with Mel warping and DCT which will be explained in the next subsections [16][37].

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j(2\pi/N)kn} \quad (10)$$

5.2.1.6 Mel-Filterbank

The Mel-filterbank is constructed in a way that represent the human ear. Equation 11 below converts Hz to a Mel scale and this gives a linear spacing below 1000 Hz and a logarithmic scaling for the frequencies above 1000 Hz seen in figure 13. To convert from Mel back to Hz can be seen in equation 12 [16][25].

$$Mel = 2595 \log_{10} \left(1 + \frac{f}{700} \right) \quad (11)$$

$$f = 700 \left(10^{\frac{Mel}{2595}} - 1 \right) \quad (12)$$

The next step is to create triangular filters that collect energy in a Mel scale spacing, 32 filter banks are used because it was proven to be the best choice [38], and that the spacing in each filter bank is approximately 120 Mel's [16]. Figure 13A shows the 32 filterbanks used and figure 13B shows the energies extracted from a single frame.

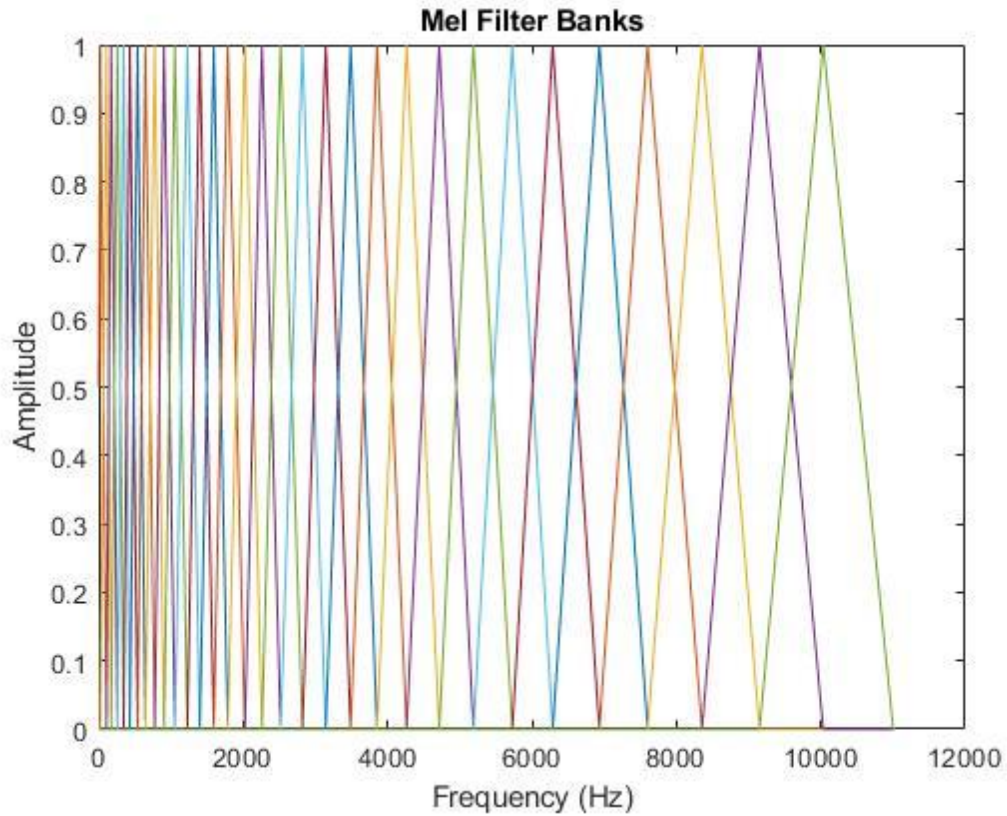


Figure 13A: The filter bank with 32 filters

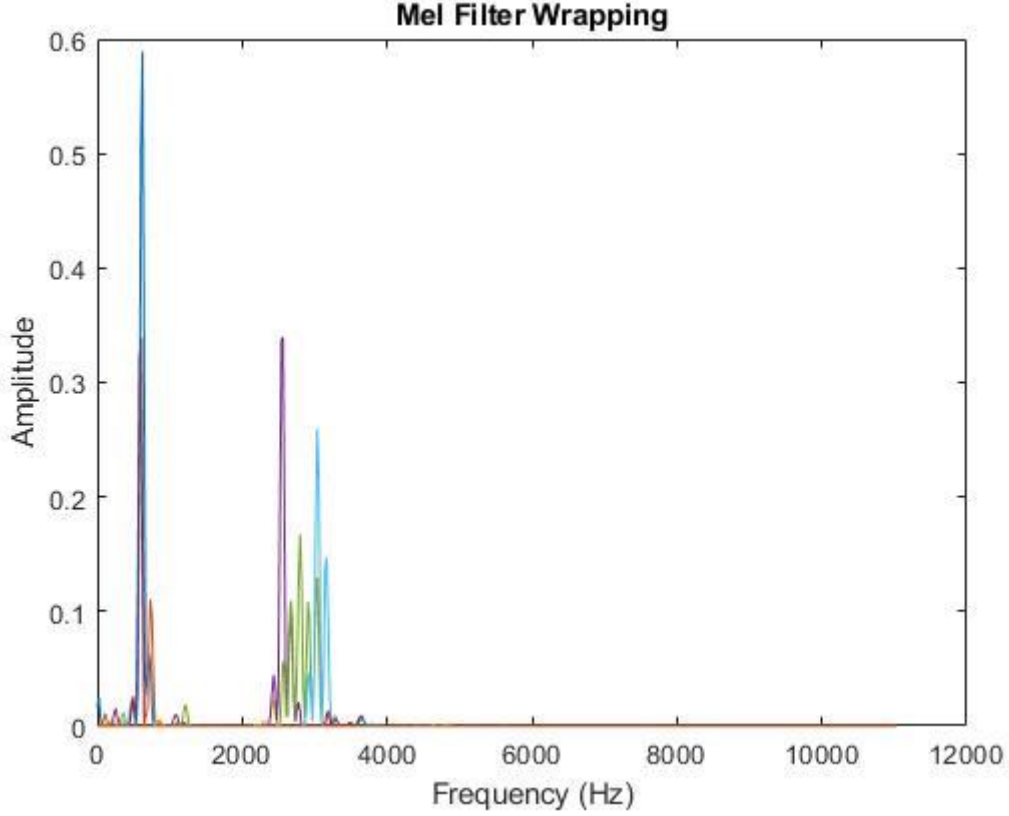


Figure 13B: The energies extracted from a single frame

5.2.1.7 Discrete Cosine Transform

DFT can be split in two parts, the real- and imaginary- part. The real part is the DCT, and the relation between the real axis and the DCT makes it an important part in speech processing [16]. Since the Mel spectrum coefficients are also real numbers then it is possible to convert them to the time domain with DCT [37]. This is the final step to obtain the MFCC features [16][25][30]. From the previous step, one can obtain 32 coefficients that is the same amount of filters used in the Mel filter bank. By obtaining coefficients number 2 to 14, this gives a value of 13 coefficients. Ignoring the first- and the higher-coefficients because it is proven to have less to no relevant information [30].

$$c_i = \sqrt{\frac{2}{N}} \sum_{j=1}^N m_j \cos\left(\frac{\pi i}{N} (j - 0.5)\right) \quad (13)$$

5.2.2 Normalization and Derivatives

Two feature normalizations will be included to improve the systems performance, CMS and feature warping.

The CMS normalization method is widely used, and is used for robustness against noise, distortion and removes the characteristics of the recording device [12][39]. There are other normalization techniques, but the CMS is more suited for this system because of the use of two microphones and the different effects they bring.

Feature warping reduces EER with 15% compared to using CMS alone [39].

Feature warping consists of mapping the observed cepstral feature distribution to a predefined normal distribution over a sliding window [21].

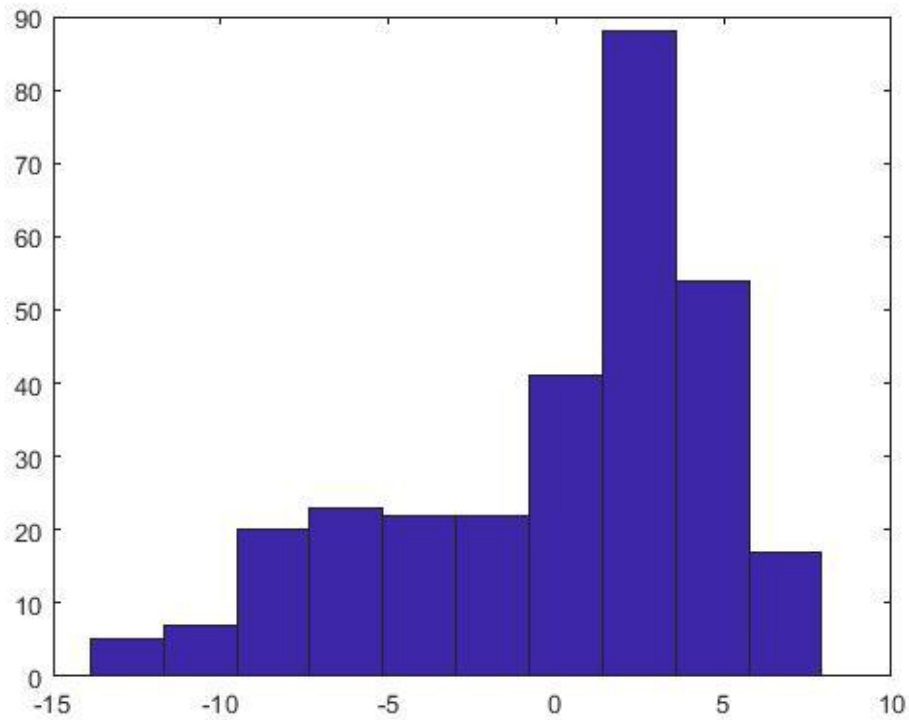


Figure 14A: Original distribution of coefficient

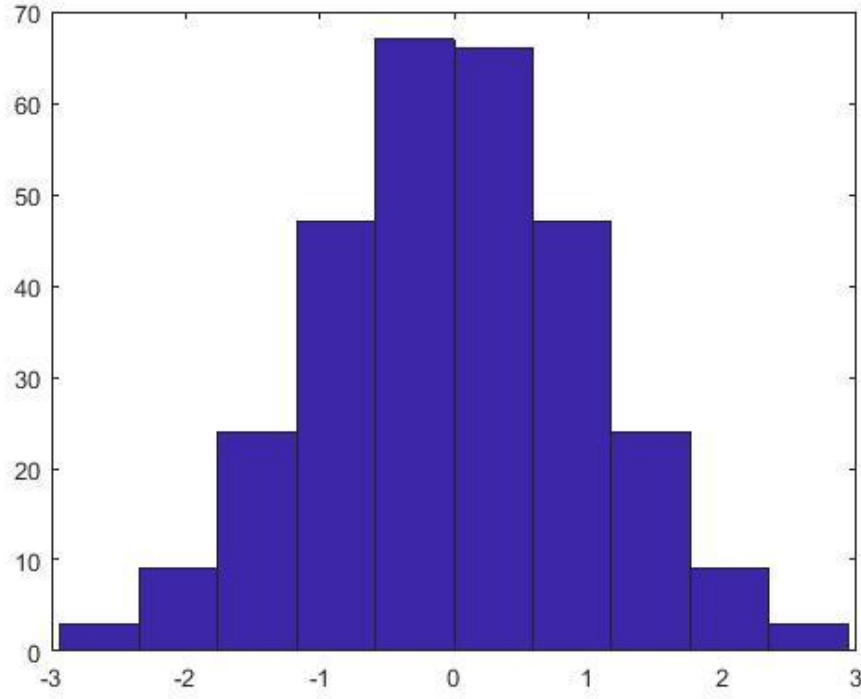


Figure 14B: Normalized distribution of coefficient

The MFCC can be further improved to give better accuracy, this is done by first compute the delta of the MFCC (DMFCCs) this is the first order derivative. It has been proven that the accuracy is even more improved by also using the second order derivative of the MFCC i.e. delta-delta MFCC (DDMFCC) [25][28].

The time derivatives of the MFCCs feature vectors enhance the performance and accuracy. The calculation is for the deltas are followed in equation 14.

$$d_t = \frac{\sum_{\theta=1}^N \theta (c_{i+\theta} - c_{i-\theta})}{2 \sum_{\theta=1}^N \theta^2} \quad (14)$$

In equation 14 the variables d_t are the delta coefficients at the time t , the variable θ is the size of the delta window [14].

5.3 The Enrollment Phase

The purpose of the enrollment phase is to create models based on the MFCCs from the feature extraction (MFCC + Delta + Delta Delta) phase. In the enrollment phase a collection of speech is obtained from 650 speakers with 70% male speakers and 30% females speakers to create the imposter model. The GMM and GMM-UBM imposter model has 1024 components and this has been proven to give a better EER compared to 2048, 512, 256 [28]. And the dimensions for this model is the amount of coefficients which are 39 (DDMFCCs). The point of focus for our research question is to create eight true speaker models (eight for GMM and eight for GMM-UBM) for two true speakers, with different amount of training data.

The duration of training data for each model is;

- Model 1 - 0.75 minutes
- Model 2 - 1.50 minutes
- Model 3 - 3.00 minutes
- Model 4 - 6.00 minutes
- Model 5 - 12.0 minutes
- Model 6 - 24.0 minutes
- Model 7 - 48.0 minutes
- Model 8 - 60.0 minutes

The results can be seen in chapter 5.4 for each model with the FP and FN along with the EER.

5.4 The Verification Phase

The Verification phase is to verify if the claimed speaker is an imposter or a true speaker. There are sixteen true speaker models for two speakers, a new collection of data is used to verify the models created and to obtain the results from it. The true speakers collect 150 new utterances with 3 seconds duration per utterance, and then ten imposter speakers that are not registered in the imposter model collect 15 utterances with 3 seconds duration per utterance. The scores can be seen in the tables in subsection 5.4.

5.5 Results

The results are presented in the tables below, with sixteen models for each speaker, and each model has a training data from 45 seconds up to 1 hour. The FN and FP in the table are based on 12 speakers and a variation of the threshold is also illustrated.

An explanation of the columns in table 2, 3, 4, 5.

- Model - The duration of the amount of training data for each model.
- Min & Max - The minimum and maximum score achieved from the speakers
- True speaker - The score of the true speaker with minimum and maximum score obtained from 150 trials.
- Imposter Speakers - The score of the imposter speakers with minimum and maximum score obtained from 150 trials.
- FP - The amount of FP if the threshold is set at the minimum value of the true speaker score (no FN)
- FN - How many times the system does not recognize the true speaker from a set of 150 trials with a threshold that is set at the maximum imposter speaker score (no FP).
- EER - Is the Equal Error Rate where the tradeoff is equal between FPR and FNR

E.g. in table 2, the column "Model One" shows the duration of the training data with 45 seconds, the "FP and FN" columns show that the FP is 18 and FN is 16, and the EER is set to 4,6667

5.5.1 The GMM-UBM Method

This subsection presents the results obtained with the GMM-UBM models.

Speaker One GMM-UBM						
Model	Min & Max Score	True Speaker	Imposter Speakers	FP (of 150) ²	FN (of 150) ³	EER (%)
One (45 sec)	Min Score	0,00870769	-0,00402882	18	16	4,6667
	Max Score	0,028357148	0,013626177			
Two (1.5 min)	Min Score	0,018041779	-0,00617504	16	11	4,0000
	Max Score	0,057024557	0,025695847			
Three (3 min)	Min Score	0,034247064	-0,011665563	19	12	3,3333
	Max Score	0,110756034	0, 048805133			
Four (6 min)	Min Score	0,062659219	-0,022923001	22	13	3,333
	Max Score	0,201937702	0,094237107			
Five (12 min)	Min Score	0,117817102	-0,033097339	15	9	1.3333
	Max Score	0,375977933	0,174932044			
Six (24 min)	Min Score	0,197142321	-0,05674105	7	11	1,3333
	Max Score	0,62512666	0,276985393			
Seven (48 min)	Min Score	0,306788539	-0,081925837	3	3	1,3333
	Max Score	0,867689006	0,365730796			
Eighth (60 min)	Min Score	0,336153946	-0,09077135	3	3	1,3333
	Max Score	0,987862643	0,395860276			

Table 2: Results for speaker one based on eight models with different amount training data with GMM-UMB modeling.

² If the threshold is set at the minimum value for the true speaker (no false negatives)

³ If the threshold is set above the maximum false speakers (no False Positives)

Speaker Two GMM-UBM						
Model	Min & Max Score	True speaker	Imposter Speaker	FP (of 150) ⁴	FN (of 150) ⁵	EER (%)
One (45 sec)	Min Score	0,006600232	-0,000970263	74	45	12,6667
	Max Score	0,031390045	0,014656359			
Two (1.5 min)	Min Score	0,008591385	-0,003094435	113	19	4,6667
	Max Score	0,06315318	0,027028485			
Three (3 min)	Min Score	0,016512423	-0,005580669	86	28	8,0000
	Max Score	0,120073816	0,046244904			
Four (6 min)	Min Score	0,019319328	-0,009219546	108	48	13,3333
	Max Score	0,235497945	0,09150491			
Five (12 min)	Min Score	0,026853545	-0,019185077	129	55	16,0000
	Max Score	0,45612731	0,187558776			
Six (24 min)	Min Score	0,124529395	-0,022079607	69	27	3,3333
	Max Score	0,549137165	0,289817779			
Seven (48 min)	Min Score	0,296393203	-0,049911876	21	4	0,6667
	Max Score	0,780212116	0,381950347			
Eighth (60 min)	Min Score	0,418886003	-0,064965486	3	1	0,6667
	Max Score	0,967036527	0,426439902			

Table 3: Results for speaker two based on eight models with different amount training data with GMM-UMB modelling.

⁴ If the threshold is set at the minimum value for the true speaker (no false negatives)

⁵ If the threshold is set above the maximum false speakers (no False Positives)

The score distribution comparison between the true speaker and imposter speakers trials from the experiment, are shown in the figures 15 below for two models (model one and model eight). These figures show the score collisions between the imposters and the true speakers and the amount of trials colliding. This gives an understanding for more data gives less collision e.g. in model eight compared to model one.

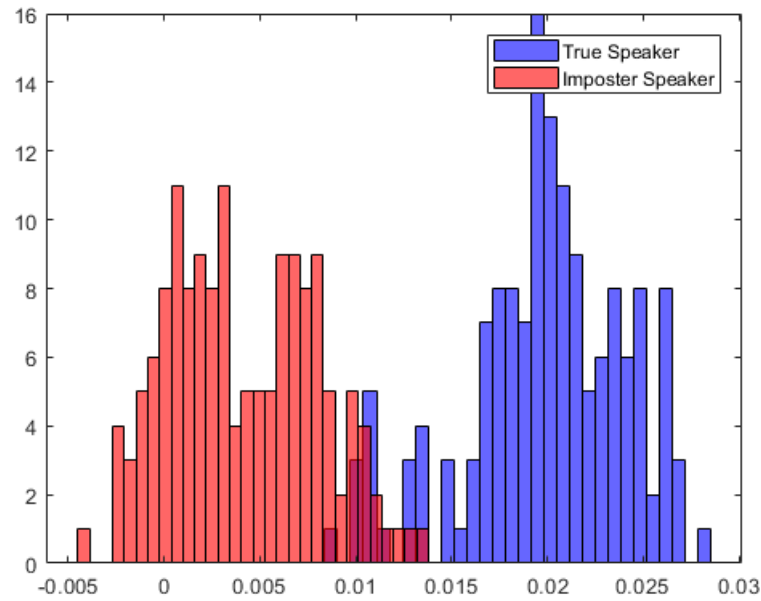


Figure 15A: Shows the score distribution for model one between the true speaker and imposter speaker (speaker one).

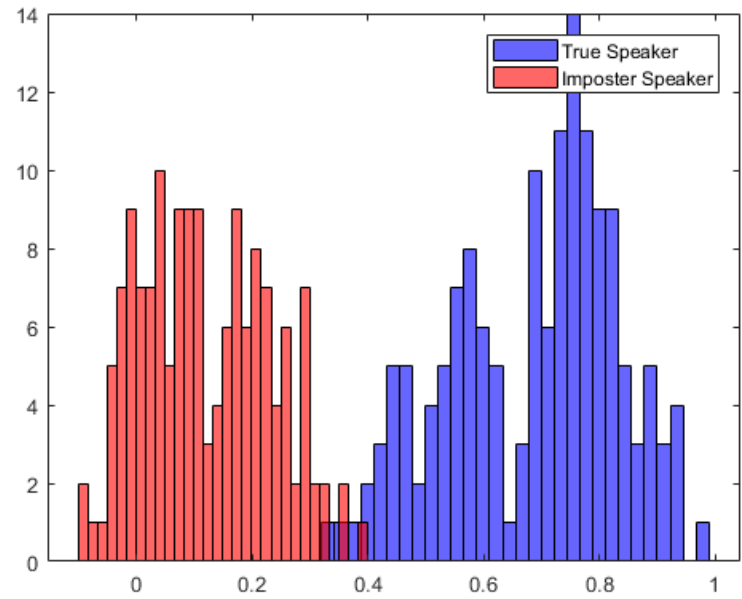


Figure 15B: Shows the score distribution for model eight between the true speaker and imposter speaker (speaker one).

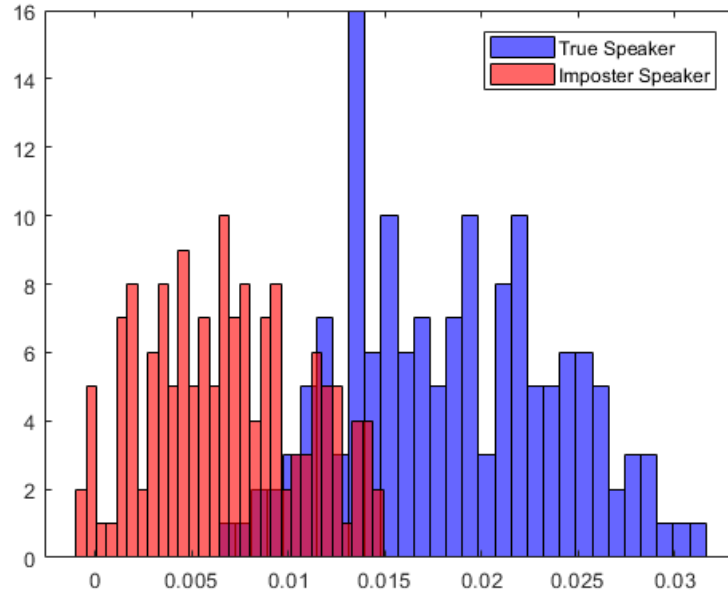


Figure 15C: Shows the score distribution for model one between the true speaker and imposter speaker (speaker two).

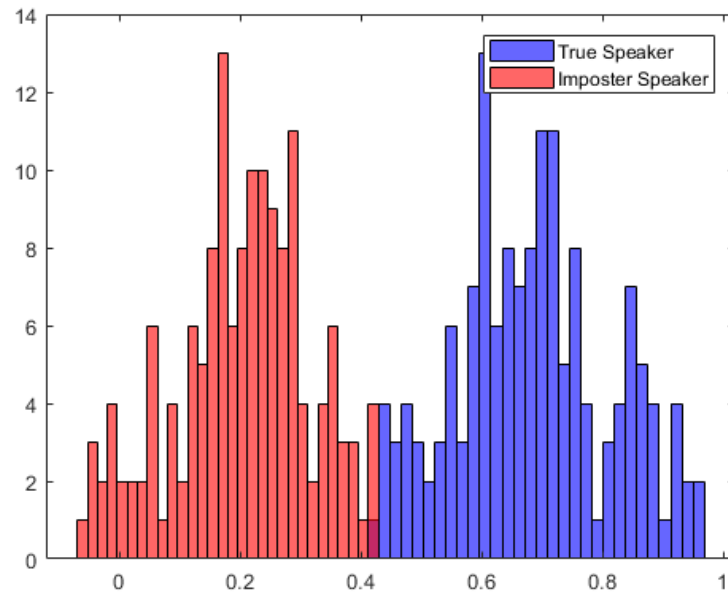


Figure 15D: Shows the score distribution for model eight between the true speaker and imposter speaker (speaker two).

The DET curves for the results are illustrated in figures 16 for the two true speakers (model one and model eight), as mentioned the threshold is varied as seen in table 2 and

3 for best FP or FN and this is dependent of the system application if there is a need for high security (e.g. door access control or turn on a lightbulb at home) or not.

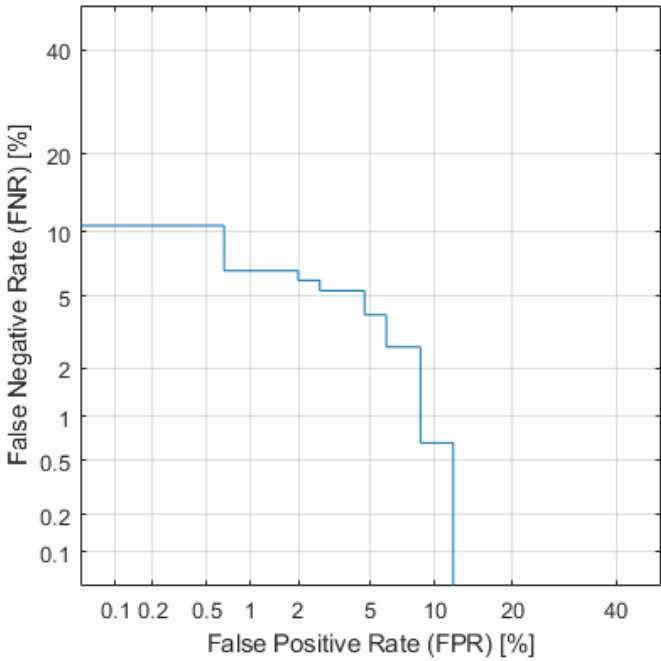


Figure 16A: Shows the DET curve for model one (speaker one).

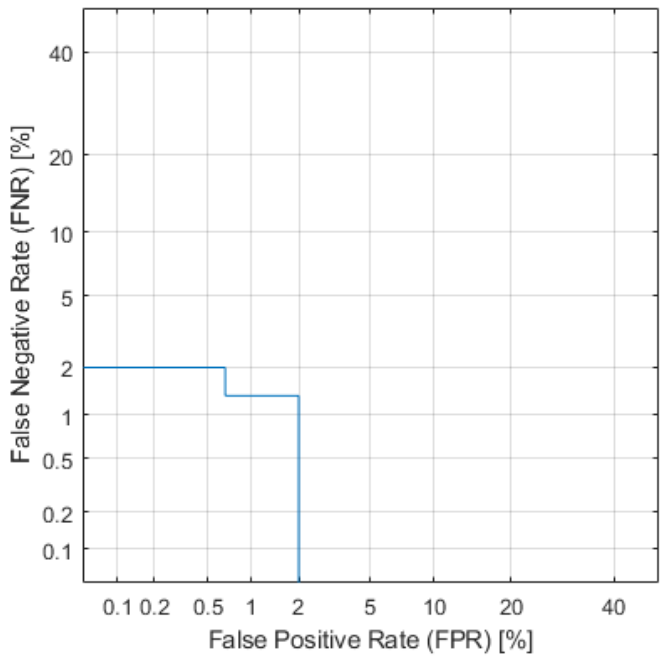


Figure 16B: Shows the DET curve for model eight (speaker one).

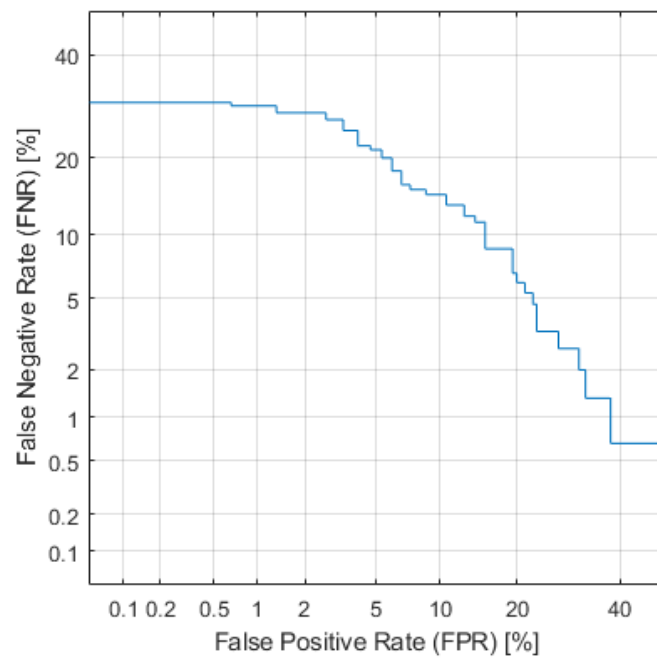


Figure 16C: Shows the DET curve for model one (speaker two).

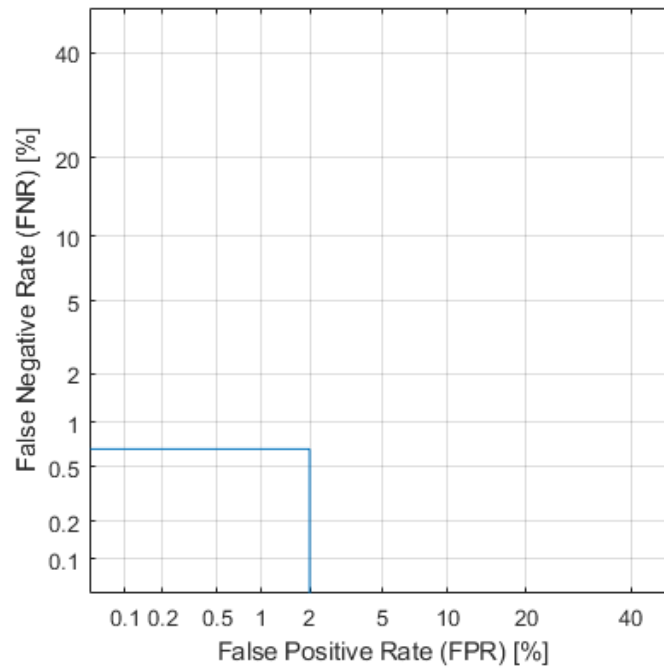


Figure 16D: Shows the DET curve for model eight (speaker two).

5.5.2 The GMM Method

This subsection presents the results obtained with the GMM models for the two true speakers.

Speaker One GMM						
Model	Min & Max Score	True speaker	Imposter Speakers	FP (of 150) ⁶	FN (of 150) ⁷	EER (%)
One (45 sec)	Min Score	-4,59543	-7,50121	44	20	6,6667
	Max Score	0,343398	-3,08241			
Two (1.5 min)	Min Score	-5,091305964	-8,32178888	69	13	4,0000
	Max Score	1,25342494	-3,163740221			
Three (3 min)	Min Score	-4,468522568	-8,210498733	61	10	4,0000
	Max Score	2,071358156	-2,324879252			
Four (6 min)	Min Score	-2,36996	-7,21947	18	7	2,6667
	Max Score	3,500626	-1,49735			
Five (12 min)	Min Score	-0,91475	-5,30245	19	8	2,6667
	Max Score	4,499395	0,429418			
Six (24 min)	Min Score	0,237479	-4,49186	13	9	2,6667
	Max Score	5,646987	1,349721			
Seven (48 min)	Min Score	0,83629	-3,96628	9	7	2,0000
	Max Score	5,958264	1,644985			
Eighth (60 min)	Min Score	0,674924	-3,7713	14	7	2,0000
	Max Score	6,381513	1,656629			

Table 4: Results for speaker one based on eight models with different amount training data with GMM modeling.

⁶ If the threshold is set at the minimum value for the true speaker (no false negatives)

⁷ If the threshold is set above the maximum false speakers (no False Positives)

Speaker Two GMM						
Model	Min & Max Score	True speaker	Imposter Speakers	FP (of 150) ⁸	FN (of 150) ⁹	EER (%)
One (45 sec)	Min Score	-6,138873088	-8,206366027	70	129	15,3333
	Max Score	-2,438541047	-3,163293335			
Two (1.5 min)	Min Score	-5,120435726	-7,40280298	62	40	8,0000
	Max Score	-1,023379218	-3,241826783			
Three (3 min)	Min Score	-5,140078025	-6,768919165	111	69	10,6667
	Max Score	0,23541799	-1,863244273			
Four (6 min)	Min Score	-4,068188027	-6,343241034	103	55	12,0000
	Max Score	1,729035451	-1,357690028			
Five (12 min)	Min Score	-3,906037689	-5,627433894	113	49	15,3333
	Max Score	3,601742494	-0,660873632			
Six (24 min)	Min Score	-0,832174144	-4,250593508	32	82	8,0000
	Max Score	3,298993342	1,290676086			
Seven (48 min)	Min Score	0,168312689	-3,418247052	24	26	4,6667
	Max Score	6,404687841	1,503379798			
Eighth (60 min)	Min Score	-0,046211077	-3,408569693	35	17	4,0000
	Max Score	6,815677411	1,365551725			

Table 5: Results for speaker one based on eight models with different amount training data with GMM modeling.

⁸ If the threshold is set at the minimum value for the true speaker (no false negatives)

⁹ If the threshold is set above the maximum false speakers (no False Positives)

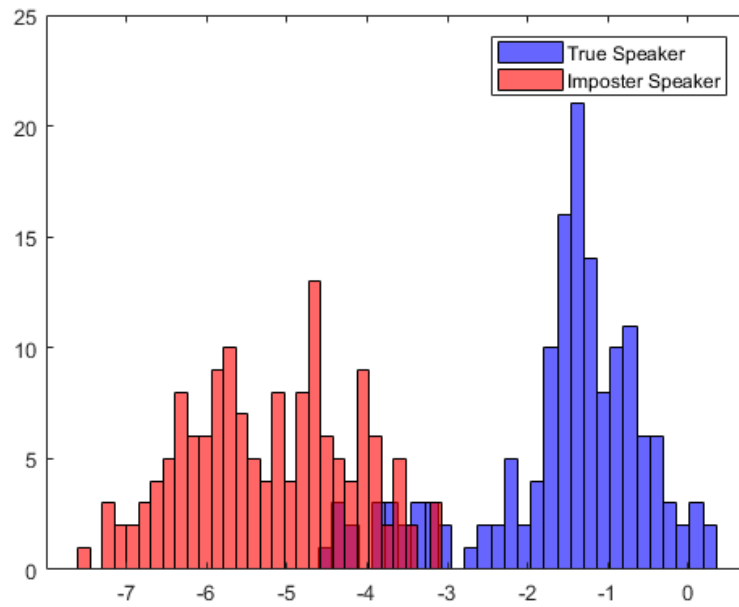


Figure 17A: Shows the score distribution for model one between the true speaker and imposter speaker (speaker one).

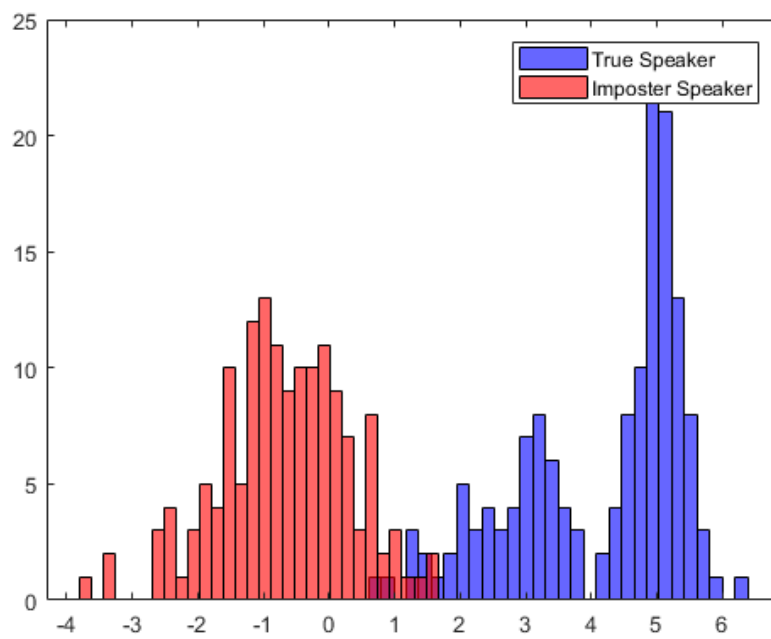


Figure 17B: Shows the score distribution for model eight between the true speaker and imposter speaker (speaker one).

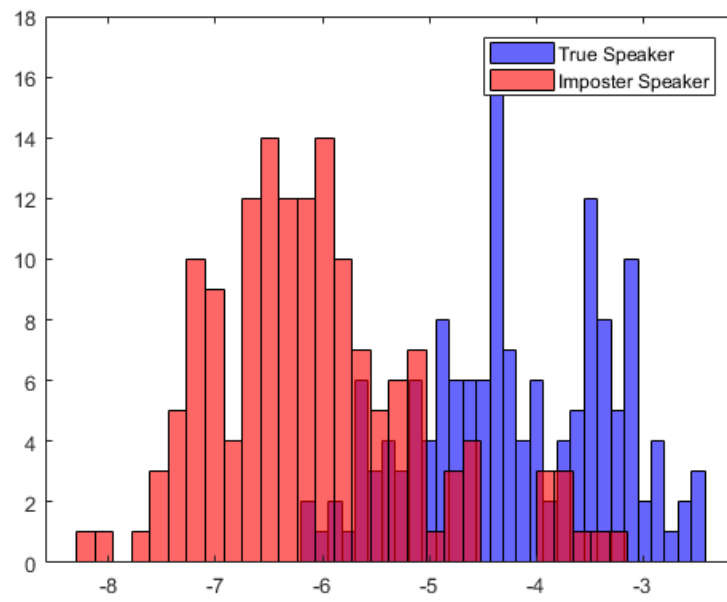


Figure 17C: Shows the score distribution for model one between the true speaker and imposter speaker (speaker two).

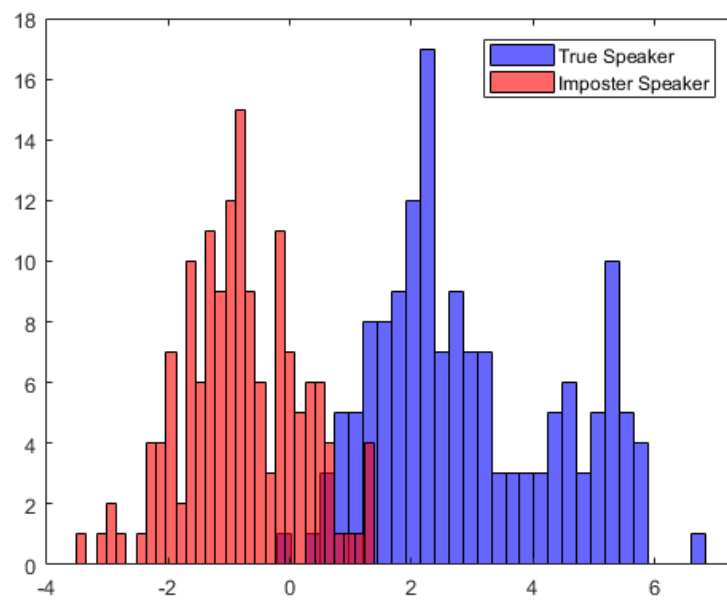


Figure 17D: Shows the score distribution for model eight between the true speaker and imposter speaker (speaker two).

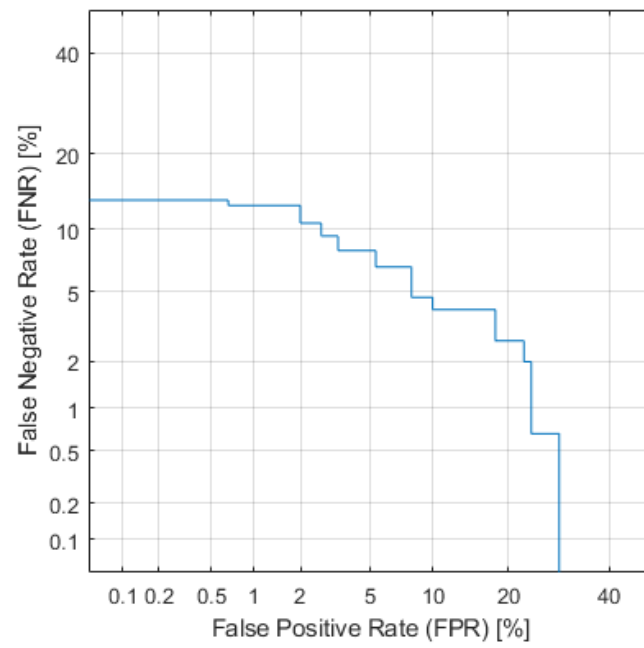


Figure 18A: Shows the DET curve for model one (speaker one).

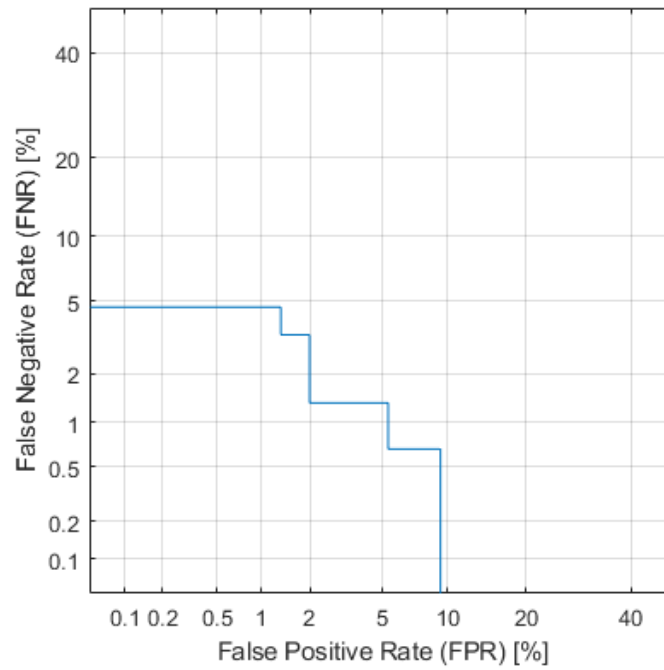


Figure 18B: Shows the DET curve for model eight (speaker one).

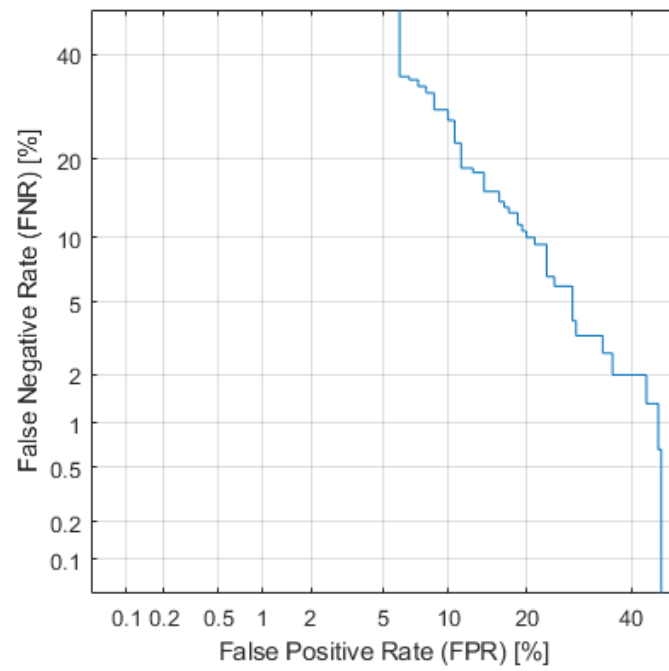


Figure 18C: Shows the DET curve for model one(speaker two).

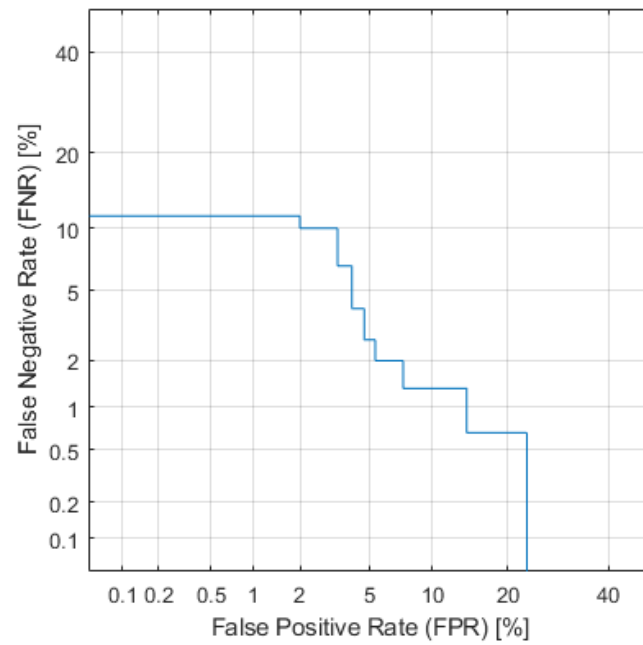


Figure 18D: Shows the DET curve for model eight (speaker two).

5.6 Analysis

The curve in figures 19 shows the EER performance of models for both GMM and GMM-UBM. The amount of training data needed shows that at 12 minutes for speaker one shows the same EER for the GMM-UBM, while for speaker two the EER reaches the lowest value at 48 minutes and is the same at 60 minutes.

By analysing the EER in figure 19A it can be seen that at least 12 min of true speaker data is needed for training the true speaker specific model, and yet this does not tell the whole truth because if we look at figure 20A, one can see that there are advantages with more training data, and for this thesis we are focusing on security. This means that when increasing the threshold as high as possible (so that zero imposters passes as true speakers) one can see that more training data (more than 12 min) will benefit the FN results. The maximum performance is reached at 48 min for our system and more than that shows no significant change to the performance. There is no advantage for increasing the training data from 48- to 60 minutes of training as seen in table 2.

While for speaker two in figure 19B shows that 12 minutes of data is not equally good with the obtained EER values. But instead at around 48 it reaches its minimum peak value.

This figures 19 below shows that with 45 seconds of training data the GMM-UBM outperforms the GMM while at approximately 6 minutes the GMM has a lower EER compared to GMM-UBM, while with 24- to 60 minutes the EER is lower for the GMM-UBM.

FN is chosen as our primary metric for evaluation of the speaker verification system performance, the reason is to be able to construct a system that provide high security against imposters and with low FN for the true speakers.

The FN for speaker one with GMM and GMM-UBM shows that the performance of the GMM-UBM is better at 48- to 60 minutes, while from 3- to 24 minutes the GMM is the better option. For speaker two the best achieved FN is at 60 minutes with GMM-UBM and the GMM only out performance the GMM-UBM at 12 minutes.

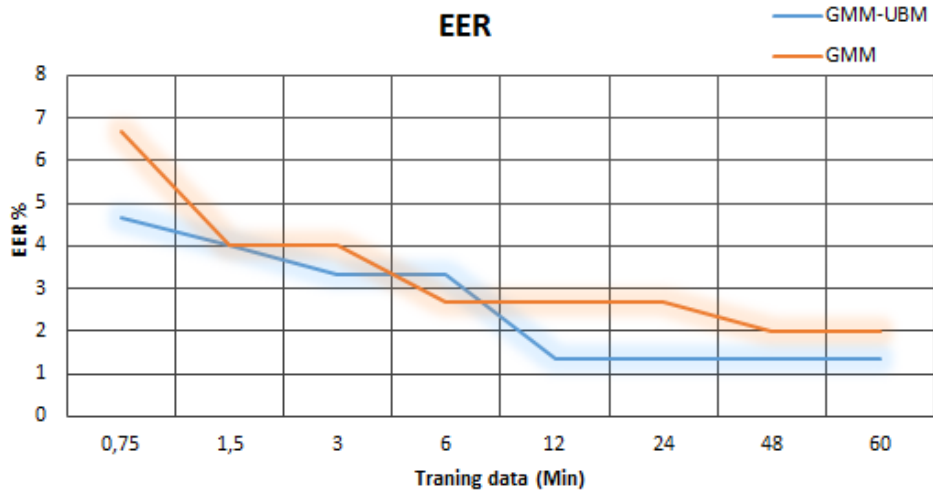


Figure 19A: The EER for the eight models for GMM and GMM-UBM (speaker one).

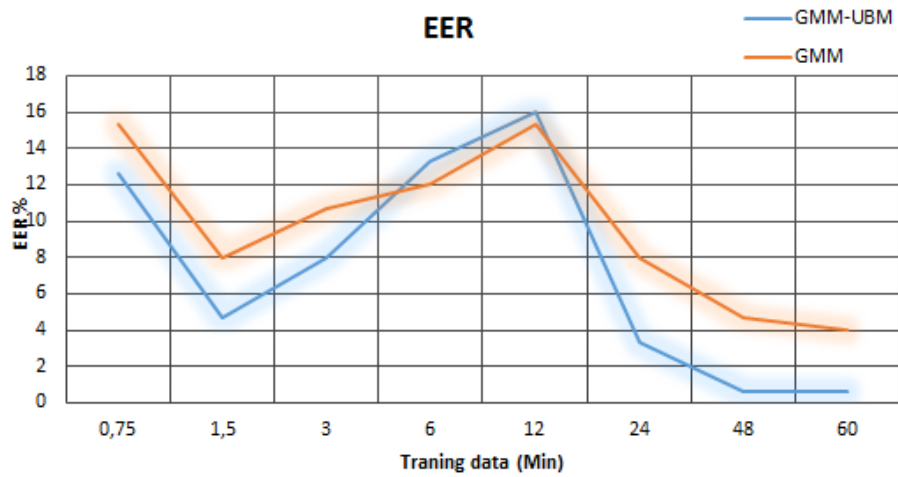


Figure 19B: The EER for the eight models for GMM and GMM-UBM (speaker two).

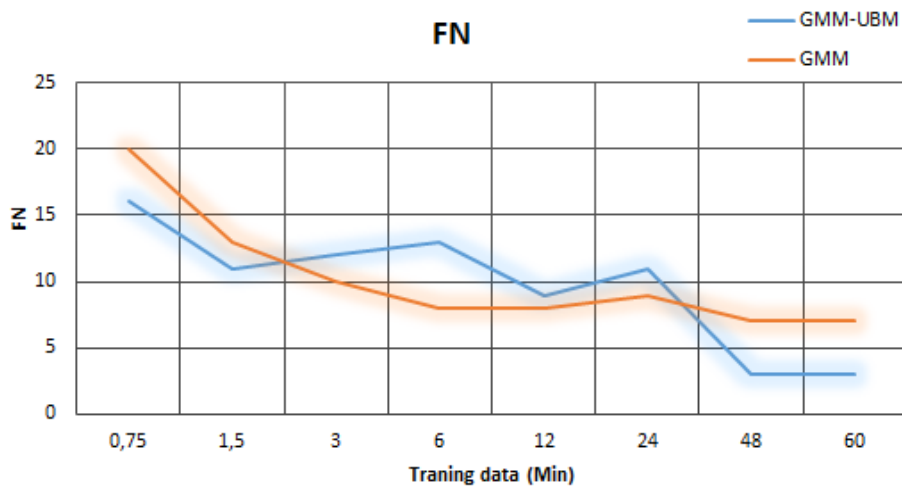


Figure 20A: The FN for the eight models for GMM and GMM-UBM (speaker one).

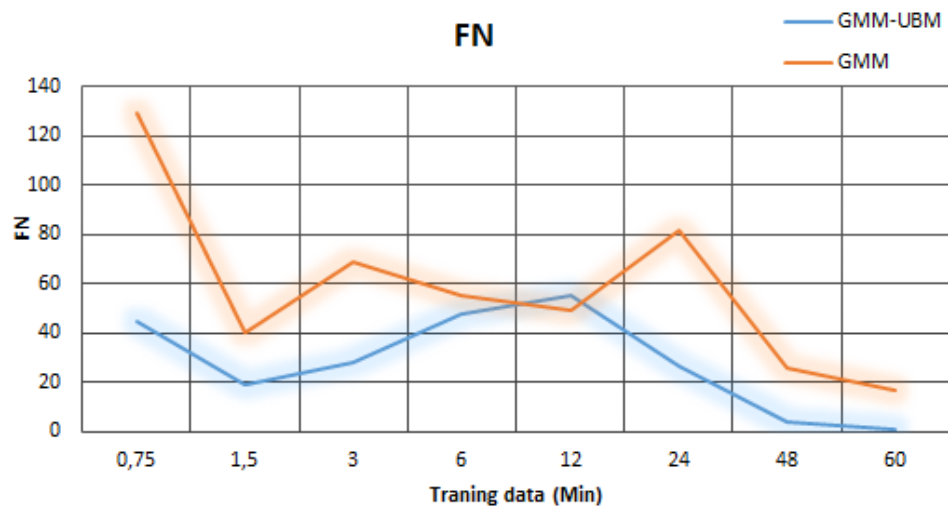


Figure 20B: The FN for the eight models for GMM and GMM-UBM (speaker two).

6 Discussion

There are no silver bullets when searching for the best performing speaker verification system, because the field is new and there is much left to be researched. This chapter will discuss and evaluate the performance and usage of the speaker verification system.

6.1 Method Discussion

The method that was used in this project can only be seen as successful, by following an iterative developmental method then it was only a given that results were to be achieved at the development (prototype) stage. Regardless of how the system architecture is build, it is always rewarding to create and develop a system as soon as possible, because there are occasions where only when the system is built it is possible to see some unexpected irregularities in the system.

By following the workflow of the systems development research methodology seen in figure 7, beginning at “construct a conceptual framework” all the way to “analyze and develop the system” can be seen as the preparation phase, where collecting as much data as possible was required in order to gain as much knowledge as possible about the system architecture before developing the system. The system apparatus used is Matlab.

6.2 The Speaker Verification System Components

The first component for a speaker verification system is the front-end. The front-end or feature extractor is essential for the system because the characteristics of the features will strongly determine the quality of the verification (pattern matching). In chapter 3 we presented a few papers with related work, and like many of these works we have decided to use the MFCC method for the feature extraction. The MFCC method provides better performance than LPCC when the environment is nosy, and it computes faster than LPCC (approximately 500%) [41][42], this makes it a more suitable feature extraction method that can be implemented in a real time system (e.g. door station, mobile banking, etc..). And to further improve the MFCC method then following Hossan. M.A., et al.s paper is needed. And along with the results obtained from Kumari, T.J., et al. that shows the use of 13 cepstral coefficients with the Delta and Delta-Delta for their system gives lower EER. Therefore, we decided to use 13 cepstral coefficients along with Delta and Delta-Delta. Furthermore, Mohamed, et al, used a window size of 32 ms and chose to use a

frame-overlap of 10ms, while we chose to have a window of 20ms and a frame-overlap of 10ms to smoothen the transition from one frame to the next even further.

By following the guidelines of *Mohamed, K.O. et al. Training Universal Background Models for Speaker Recognition* and *Taufiq, H. et al. A Study on Universal Background Model Training in Speaker Verification* we decided to use the GMM and GMM-UBM method for the second component which is the enrollment phase and verification phase.

For the GMM-UBM the chosen gaussians components were 1024 for us, Mohamed, et al., and Kumari, T.J., et al. which has been proven to be the better choice.

6.3 Speaker Verification System Performance

The system performance is tightly coupled with the enrollment phase and the GMM-UBM as seen in table 2 for speaker one. To answer the research questions in section 1.2 we decided to make eight models with a varied training data on each model from 0.75 min to 60 min. The results shows an great decrease in EER depending on how much training data is used, this is illustrated in figure 19A. A point that was noticed in these results is what *Taufiq, H. et al.* said in *A Study on Universal Background Model Training in Speaker Verification*, that there is no concrete evidence that says the more data used the better the system will perform. This can be seen in our results with figure 19A where the EER is at its lowest already at 12 minutes of data. While if you consider a more secure system then looking at figure 20A then it is possible to see that after 48 min of data the system performance is more than sufficient and more data will not necessarily improve the performance of the system to be much better.

While for speaker two the GMM-UBM models performance is slightly different when looking at the EER, one thing that can be seen is that at 48 minutes the performance reaches the minimum EER. While for figure 20B at 48- and 60 minutes there is a slight difference of approximately 1.3% FNR.

Kumari, T.J., et al. said in their study that the EER obtained with GMM is approximately 35%, while with GMM-UBM the EER is approximately 25%, and they also say that the GMM-UBM is the better option when having a true speaker training data of less than 15 seconds. It is true that the result obtained in this research shows that the GMM-UBM outperforms the GMM when looking at the EER values, but when considering the security aspects of the system then another point of focus is the FN for the speaker models. The

FN shows in turn different results that cannot be seen from the EER when considering security, and this can be seen below.

Speaker One:

- GMM outperformed GMM-UBM 1 of 8 models (EER measurement)
- GMM outperformed GMM-UBM 4 of 8 models (FN measurement)

Speaker Two:

- GMM outperformed GMM-UBM 2 of 8 models (EER measurement)
- GMM outperformed GMM-UBM 1 of 8 models (FN measurement)

There is no pattern to be seen between the EER and FN, while Kumari, T.J. states that the GMM-UBM is the better choice with the EER result obtained, they don't consider the security aspect and what that would result in. Although one pattern that can be seen is that approximately in 6- to 12 minutes the GMM gave better performance (both with EER and FN).

7 Conclusion

A speaker verification system can be designed in many ways, both the front-end and the back-end, when developing a speaker verification system then it is important to consider the environment, requirements and limitations that exist which will ease the choice of the right implementation, and in turn give the best performance.

In this section, we will answer the main point of importance to the research question in section 1.2 with the requirements on security and the limitation of having up to 1 hour of data.

7.1 Answering the Research Question

The research questions mentioned in section 1.2, will be answered in a chronological order to reach a conclusion to which of these amount of data options is more secure for the GMM-UBM (0.5, 1.5, 3, 6, 12, 24, 48, and 60 minutes) and which classifier (GMM and GMM-UBM) is more secure for these amounts of data collected. this can be seen in the subsections below.

7.1.1 How does the GMM-UBM speaker verification system perform based on different amount of data collection for the true speaker model?

Two evaluation metrics were used to evaluate the system performance, these have a different meaning for the biometric system. The first evaluation metric is the EER and which shows that the model containing 48 minutes gave the best results. And the second metric is the FN which is related to security when FP is at zero, here the results also shows that 48 minutes gave the best performance and that there were little to no difference in the performance compared to the 60 minutes model.

Conclusion:

The models reach the optimal performance at 48 minutes and there is no difference in performance when increasing the data to 1 hour for the first speaker while for the second speaker there was a small increase in performance with 1 hour of training data.

7.1.2 How does the GMM speaker verification system perform compared to the GMM-UBM, based on different amount of data collection for the true speaker model?

The results on which of the two options mentioned (GMM and GMM-UBM) is the most efficient depends on the amount of training data, and how secure the system should be e.g. looking at the FN and FP of the system. This is presented in chapter 5 in figures 19 and 20. The EER metrics shows that the GMM-UBM outperformed GMM with 13 out of 16 models, and even for the FN metrics where 11 out of 16 models were better with GMM-UBM.

Conclusion:

The main difference is in the amount of training data, when choosing the most secure option out of these models then GMM-UBM is the better choice.

7.2 Further Work

A future work could be to use LPCC for the feature extraction instead of MFCC and how would the results differ from this research, and compare GMM-SVM to GMM and GMM-UBM.

Another interesting future work could be an extension of Kumari, T.J., research that concluded that LPCC was the better option compared to MFCC when computing with less than 15 seconds of training data, but would the results change and favour the MFCC if they would use more training data such as up to 1 hour like in our case?

7.3 Contribution of this Thesis

In this thesis, the primary contributions for this work has been to research on how much data is necessary for training a secure speaker verification system and the differences between two classifiers used in the area of speaker recognition. This gives the readers a broader knowledge on the advantages that the amount of training data has and shows the advantages and challenges that the GMM and GMM-UBM brings with different amount of data when considering to implement a speaker verification system.

References

- [1]Pato, J. N., & Millett, L. I. (2010). Biometric Recognition: Challenges and Opportunities, Whither Biometrics Committee, National Research Council of the NSA.
- [2]Jain, A. K., Ross, A., & Prabhakar, S. (2004). An introduction to biometric recognition. *IEEE Transactions on circuits and systems for video technology*, 14(1), 4-20.
- [3]Travieso-González, C. M., & Alonso-Hernández, J. (2011). Advances in Nonlinear Speech Processing: International Conference on Nonlinear Speech Processing, No-LISP 2011, Las Palmas de Gran Canaria, Spain, November 7-9, 2011. *Proceedings. Lecture Notes in Computer Science/Lecture Notes in Artificial Intelligence*. Springer.
- [4]Robertson, B. (2016). Q: How Does Speech-Recognition Software Work?. *Science and Children*, 54(3), 64.
- [5]Pieraccini, R. (2012). *The voice in the machine: building computers that understand speech*. MIT Press.
- [6]Caranica, A., Cucu, H., Burileanu, C., Portet, F., & Vacher, M. (2017, July). Speech recognition results for voice-controlled assistive applications. In *Speech Technology and Human-Computer Dialogue (SpeD), 2017 International Conference on*(pp. 1-8). IEEE.
- [7]Leu, F. Y., & Lin, G. L. (2017, March). An MFCC-based speaker identification system. In *Advanced Information Networking and Applications (AINA), 2017 IEEE 31st International Conference on* (pp. 1055-1062). IEEE.
- [8]Tsujikawa, M., Nishikawa, T., & Matsui, T. (2017, October). I-vector-based speaker identification with extremely short utterances for both training and testing. In *Consumer Electronics (GCCE), 2017 IEEE 6th Global Conference on* (pp. 1-4). IEEE.
- [9]Dumpala, S. H., & Kopparapu, S. K. (2017, May). Improved speaker recognition system for stressed speech using deep neural networks. In *Neural Networks (IJCNN), 2017 International Joint Conference on* (pp. 1257-1264). IEEE.
- [10]Shin, D. G., & Jun, M. S. (2015, July). Home IoT device certification through speaker recognition. In *Advanced Communication Technology (ICACT), 2015 17th International Conference on* (pp. 600-603). IEEE.
- [11]Lin, C. Y., Song, K. T., Chen, Y. W., Chien, S. C., Chen, S. H., Chiang, C. Y., ... & Liu, T. J. (2012, October). User identification design by fusion of face recognition and speaker recognition. In *Control, Automation and Systems (ICCAS), 2012 12th International Conference on* (pp. 1480-1485). IEEE.
- [12]Snyder, D., Ghahremani, P., Povey, D., Garcia-Romero, D., Carmiel, Y., & Khudanpur, S. (2016, December). Deep neural network-based speaker embeddings for end-to-end speaker verification. In *Spoken Language Technology Workshop (SLT), 2016 IEEE* (pp. 165-170). IEEE.
- [13]Hasan, T., & Hansen, J. H. (2011). A study on universal background model training in speaker verification. *IEEE Transactions on Audio, Speech, and Language Processing*, 19(7), 1890-1899.
- [14]Reynolds, D. A., Quatieri, T. F., & Dunn, R. B. (2000). Speaker verification using adapted Gaussian mixture models. *Digital signal processing*, 10(1-3), 19-41.

- [15]Sigmund, M. (2007). VOICE ANALYSIS AND RECOGNITION
- [16]Beigi, H. (2011). *Fundamentals of speaker recognition*. Springer Science & Business Media.
- [17]McClellan, J. H., Schafer, R. W., & Yoder, M. A. (2003). *Signal processing first*. Upper Saddle River, NJ: Pearson education.
- [18]Tan, L. (2008). *Digital signal processing. [electronic resource] : fundamentals and applications*. Amsterdam ; Boston : Academic Press, c2008.
- [19]Smith, S. W. (1997). The scientist and engineer's guide to digital signal processing.
- [20]Hanchate, D. B., Nalawade, M., Pawar, M., Pophale, V., & Maurya, P. K. (2010, April). Vocal digit recognition using artificial neural network. In *Computer Engineering and Technology (ICCET), 2010 2nd International Conference on* (Vol. 6, pp. V6-88). IEEE.
- [21]Shriberg, E. (2007). Higher-level features in speaker recognition. In *Speaker Classification I* (pp. 241-259). Springer, Berlin, Heidelberg.
- [22]Parker, M. (2017). *Digital Signal Processing 101: Everything you need to know to get started*. Newnes.
- [23]Bisina, K. V., & Azeez, M. A. (2017, June). Optimized estimation of power spectral density. In *Intelligent Computing and Control Systems (ICICCS), 2017 International Conference on* (pp. 871-875). IEEE.
- [24]Löfgren, P. (n.d.). *VoiceR, A System for Speaker Recognition using Gaussian Mixture Model*(Unpublished master's thesis). Kungliga Tekniska högskolan.
- [25]Hossan, M. A., Memon, S., & Gregory, M. A. (2010, December). A novel approach for MFCC feature extraction. In *Signal Processing and Communication Systems (ICSPCS), 2010 4th International Conference on* (pp. 1-5). IEEE.
- [26]Al-Kaltakchi, M. T., Woo, W. L., Dlay, S. S., & Chambers, J. A. (2016, March). Study of fusion strategies and exploiting the combination of MFCC and PNCC features for robust biometric speaker identification. In *Biometrics and Forensics (IWBIF), 2016 4th International Workshop on* (pp. 1-6). IEEE.
- [27]Najar, F., Bourouis, S., Bouguila, N., & Belghith, S. (2017, October). A comparison between different gaussian-based mixture models. In *Computer Systems and Applications (AICCSA), 2017 IEEE/ACS 14th International Conference on*(pp. 704-708). IEEE.
- [28]Kumari, T. J., & Jayanna, H. S. (2014, December). Comparison of LPCC and MFCC features and GMM and GMM-UBM modeling for limited data speaker verification. In *Computational Intelligence and Computing Research (ICCIC), 2014 IEEE International Conference on* (pp. 1-6). IEEE.
- [29]Wu, T. Y., Lu, L., Chen, K., & Zhang, H. (2003, January). Universal Background Models for Real-time Speaker Change Detection. In *MMM* (pp. 135-149).
- [30]Zheng, F., Zhang, G., & Song, Z. (2001). Comparison of different implementations of MFCC. *Journal of Computer Science and Technology*, 16(6), 582-589.

- [31]Omar, M. K., & Pelecanos, J. W. (2010, June). Training Universal Background Models for Speaker Recognition. In *Odyssey* (p. 10).
- [32]Pelecanos, J., & Sridharan, S. (2001). Feature warping for robust speaker verification.
- [33]Nunamaker Jr, J. F., Chen, M., & Purdin, T. D. (1990). Systems development in information systems research. *Journal of management information systems*, 7(3), 89-106.
- [34]Bouziane, A., Kadi, H., Hourri, S., & Kharroubi, J. (2016, October). An open and free speech corpus for speaker recognition: The FSCSR speech corpus. In *Intelligent Systems: Theories and Applications (SITA), 2016 11th International Conference on* (pp. 1-5). IEEE.
- [35]Muda, L., Begam, M., & Elamvazuthi, I. (2010). Voice recognition algorithms using mel frequency cepstral coefficient (MFCC) and dynamic time warping (DTW) techniques. *arXiv preprint arXiv:1003.4083*.
- [36]Kinnunen, T. (2003). Spectral features for automatic text-independent speaker recognition. *Licentiate's Thesis*.
- [37]Bhattacharai, K., Prasad, P. W. C., Alsadoon, A., Pham, L., & Elchouemi, A. (2017, April). Experiments on the MFCC application in speaker recognition using Matlab. In *Information Science and Technology (ICIST), 2017 Seventh International Conference on* (pp. 32-37). IEEE.
- [38]Bhattacharai, K., Prasad, P. W. C., Alsadoon, A., Pham, L., & Elchouemi, A. (2017, April). Experiments on the MFCC application in speaker recognition using Matlab. In *Information Science and Technology (ICIST), 2017 Seventh International Conference on* (pp. 32-37). IEEE.
- [39]Barras, C., & Gauvain, J. L. (2003, April). Feature and score normalization for speaker verification of cellular data. In *Acoustics, Speech, and Signal Processing, 2003. Proceedings.(ICASSP'03). 2003 IEEE International Conference on* (Vol. 2, pp. II-49). IEEE.
- [40]Salla, R., Wilhelmiina, H., Sari, K., Mikaela, M., Pekka, M., & Jaakko, M. (2018). Evaluation of the confusion matrix method in the validation of an automated system for measuring feeding behaviour of cattle. *Behavioural processes*.
- [41]Misra, S., Das, T., Saha, P., Baruah, U., & Laskar, R. H. (2015, March). Comparison of MFCC and LPCC for a fixed phrase speaker verification system, time complexity and failure analysis. In *Circuit, Power and Computing Technologies (ICCPCT), 2015 International Conference on* (pp. 1-4). IEEE.
- [42]Bhattacharjee, U. (2013). A comparative study of LPCC and MFCC features for the recognition of Assamese phonemes. *International Journal of Engineering Research and Technology*, 2(1).